Near-term Quantum Computing — lecture notes

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2026-10-22

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1 Platforms, gates and measurement

On paper, a quantum algorithm is a circuit: gates on ideal qubits, followed by measurements. A real quantum computer is a physical device, whose physics decides which operations it can perform, between which qubits, how fast and how reliably. In this chapter, we look at this physical side and at what it means for the circuits we can run.

By the end of the chapter, you should be able to explain what a quantum computer needs from its qubits and how superconducting circuits and trapped ions provide it; why devices have different native gates, why Z rotations are usually free, and what limited connectivity costs; and how many shots an expectation value needs and how long they take.

1.1 Noisy qubits and fault tolerance

Every physical qubit is imperfect. Each gate has a small chance of going wrong, the stored quantum state degrades even when nothing is done to it, and measurements sometimes return the wrong result. These errors add up. If each gate fails with probability 0.1%, a circuit of a thousand gates runs without any error only about 37% of the time, since $0.999^{1000} \approx e^{-1}$. Noise therefore limits the size of the circuits that a device can run usefully.

Preskill called the devices of this era *NISQ* devices, for *noisy intermediate-scale quantum* (Preskill 2018). *Intermediate scale* means tens to thousands of qubits, enough that many of their circuits are hard to simulate on a classical computer, but far from the millions of qubits that the best-known quantum algorithms need. *Noisy* means that their errors are not corrected.

Getting useful results from a NISQ device means working within the limit set by noise. Circuits are kept short, so that errors have little chance to build up. The quantities to keep small are the circuit's *depth*, the number of layers of gates that must run one after another, and its number of two-qubit gates, the least reliable operations on current devices (Section 1.5). Each run of a circuit returns a single random outcome, so every quantity is estimated from many repetitions, called shots (Section 1.4.2). And *error mitigation* reduces the effect of the remaining noise on these estimates, without correcting the errors themselves, by running extra or modified circuits and combining their results. Error mitigation is not the same as error correction: it only works if the circuit does not accumulate too much error in the first place, and its price is running more shots. We come back to it later in the course.

Correcting errors is possible in principle. *Quantum error correction* stores each qubit of information redundantly in many *physical qubits*, which together form one *logical qubit*. Repeated checks on the physical qubits reveal errors without disturbing the stored information, so that the errors can be undone. This only helps if the physical qubits are good enough. Below a certain error rate per operation, the *threshold*, larger codes give lower logical error rates, as low as a computation requires, and the computer is called *fault tolerant*.

Fault tolerance is expensive. Useful algorithms are estimated to need of order a thousand physical qubits for every logical qubit, and around a million physical qubits in total (Fowler et al. 2012; Gidney 2025). Experiments are now crossing the threshold: in 2024, a superconducting device ran an error-correcting code whose logical error rate halved each time the code was made larger (Google Quantum AI and Collaborators 2025). Such experiments use a whole device to protect a single logical qubit, however, and useful computations still run on physical qubits. We come back to the stage in between, *early fault tolerance*, at the end of the course.

The cost of error correction is not only an increase in the number of physical qubits. A code protects stored information, but every gate on logical qubits must also be carried out without spreading errors, and only a small set of gates can be done this way at a low cost. Every other gate has to be built from this set, so the operations that are cheap on a NISQ device can be expensive on a fault-tolerant one, and the reverse. In the most studied codes, a CNOT between two logical qubits is relatively cheap. A single-qubit rotation by an arbitrary angle, which a NISQ device performs with a single pulse (Section 1.3.2), must instead be approximated by a sequence of tens to about a hundred protected gates, some of which need a costly extra procedure of their own (Ross and Selinger 2016; Litinski 2019). The cost of a NISQ circuit is dominated by its depth and its two-qubit gates, while the cost of a fault-tolerant circuit is often dominated by its number of rotations. Table 1.1 summarises the differences.

Table 1.1: NISQ and fault-tolerant quantum computing compared.

	NISQ	Fault tolerant
Qubits	physical qubits	logical qubits, each encoded in many physical qubits
Errors	uncorrected; they accumulate with every operation	corrected, if the physical error rates are below the threshold
Circuit size	limited by noise	limited by the number of qubits and the run time
Cheap operations	single-qubit rotations at any angle	CNOT and a few other gates
Expensive operations	two-qubit gates	single-qubit rotations at arbitrary angles
Cost of a circuit	mainly its depth and number of two-qubit gates	mainly its number of rotations

	NISQ	Fault tolerant
Dealing with noise	short circuits, many shots, error mitigation	error correction
Status	available today	first demonstrations; useful machines still to be built

This course focuses on NISQ devices. We will learn how noise affects their results, how error mitigation can reduce its effect, and which algorithms can be run on today's devices.

1.2 Physical qubits

This section is a high-level introduction to the most common designs of two kinds of hardware, superconducting circuits and trapped ions. Quantum hardware is a very active field of research, and there are many variations on everything we describe here. The device physics is descriptive, and you are not expected to derive it. What matters later are its consequences for gates, connectivity and speed. Terms we leave unexplained are in the glossary; on the web version, hover over an underlined term to see its definition.

1.2.1 Requirements for a quantum computer

A qubit is a quantum system with two states, $|0\rangle$ and $|1\rangle$, that we can control. Quantum systems such as atoms and small electrical circuits can only take certain values of energy, their *energy levels*, and they usually have many more than two. Most qubits are therefore the two lowest energy levels of a larger system. The other levels are still there, and a qubit can end up in them by accident.

A universal, *digital* quantum computer is built by assembling many physical qubits and controlling them individually. DiVincenzo listed what this requires (DiVincenzo 2000):

- a scalable system of well-characterised qubits: their levels and interactions are known, and more qubits can be added;
- a way to prepare the qubits in a known state, usually $|0 \cdots 0\rangle$;
- a universal set of gates, usually rotations of single qubits together with an interaction between two qubits;
- a way to measure individual qubits;
- *coherence times*, the times for which the qubits keep their quantum state, that are long compared with the duration of a gate.

These are the *DiVincenzo criteria*. Two further criteria concern sending qubits between distant devices, which we will not need.

One property of measurement matters well before Section 1.4, where readout is described in detail. On the platforms in this chapter, a measurement asks each qubit whether it is in $|0\rangle$ or in $|1\rangle$: it projects the qubit onto one of these two states and returns one bit. This is a measurement in the *computational basis*, also called the Z basis, since $|0\rangle$ and $|1\rangle$ are the eigenstates of Z. To measure in another basis, we first rotate that basis onto the computational one.

The requirements pull in opposite directions. A qubit keeps its state longest when it is well isolated from its surroundings, since every interaction with the environment disturbs it. Yet to apply gates and to measure, we have to interact with it, strongly and quickly. Every platform is a compromise between isolation and control, and this compromise explains many of the numbers in this chapter.

1.2.2 Energy levels, frequencies and pulses

Quantum mechanics links energy and frequency. An energy difference E between two levels corresponds to a frequency $f = E/h$, where h is Planck's constant, and radiation at that frequency drives transitions between the two levels, while radiation at other frequencies hardly affects them. This selectivity is called *resonance*, and on most platforms it is how qubits are controlled.

A *pulse* is a short burst of radiation at a chosen frequency, the carrier frequency, switched on and off smoothly. Its shape over time, the *envelope*, is often close to a Gaussian (Figure 1.1 a). Superconducting qubits are driven by microwave pulses, trapped ions by laser or microwave pulses. A qubit with transition frequency f_q oscillates at that frequency too, and in a simplified picture, the effect of a pulse on the qubit is the product of the pulse with this oscillation, added up over time. If the two frequencies are equal, the product keeps the same sign, and its effect builds up as long as the pulse lasts (Figure 1.1, left, b and c). The result is a rotation of the qubit, by an angle proportional to the area under the envelope. If the frequencies differ, the product alternates in sign, its contributions cancel, and the qubit is left almost unchanged. A pulse at the frequency of one qubit therefore acts on that qubit, and hardly on qubits with other frequencies. Section 1.3.2 shows how the shape and timing of a pulse set the rotation.

1.2.3 Superconducting qubits

A superconducting qubit is a small electrical circuit on a chip. The chip is *fabricated* like a conventional microchip: in a clean room, thin layers of superconducting metal are deposited on it and etched into shape (Figure 1.3). Once cooled, its metal becomes superconducting. Current then flows without resistance, so no energy is lost as heat, and the circuit behaves as a quantum system with discrete energy levels; we treat this part as a black box. A simple circuit

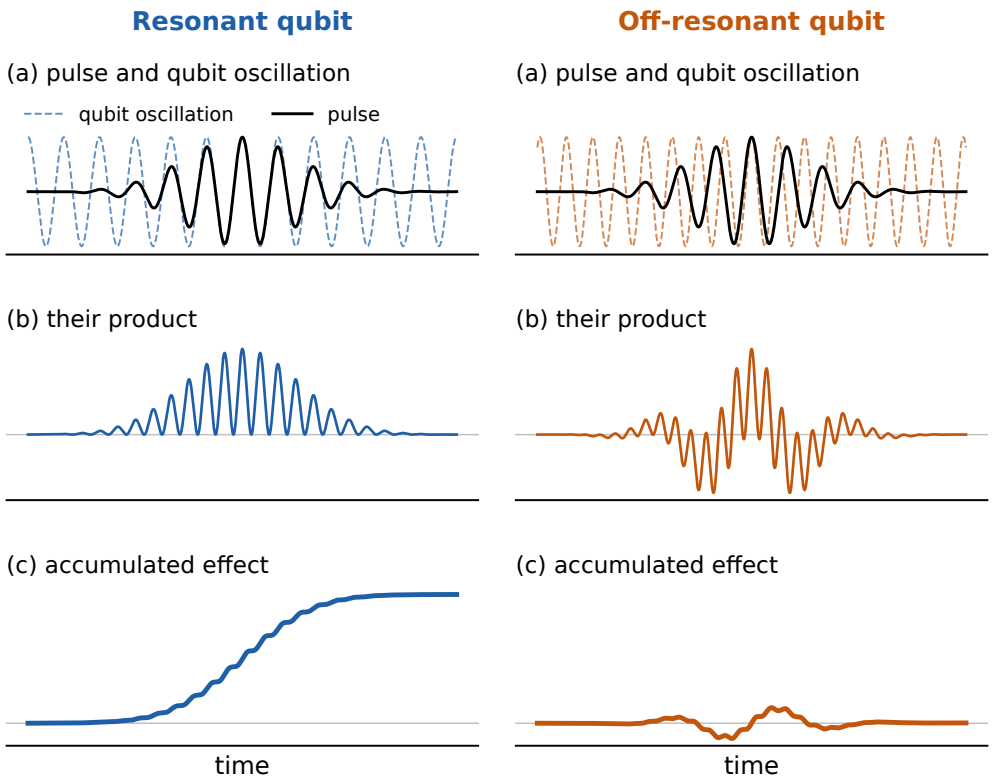


Figure 1.1: The same Gaussian pulse acting on a qubit at the carrier frequency (left) and on a qubit at a 35% higher frequency (right). (a) The pulse, with the qubit's own oscillation dashed; (b) the product of the two; (c) its running integral. The effect on the resonant qubit builds up to a value set by the area under the envelope; for the off-resonant qubit, it averages out to almost zero. The frequencies are scaled down to show the oscillations: a superconducting qubit oscillates a hundred times or more during a pulse, so that a much smaller difference in frequency is enough.

of this kind has evenly spaced levels, like a mass on a spring (Figure 1.2, left). That makes it useless as a qubit, because a pulse at the frequency f_{01} of the $|0\rangle \to |1\rangle$ transition is just as resonant with $|1\rangle \to |2\rangle$, $|2\rangle \to |3\rangle$ and so on. A circuit element called a *Josephson junction* makes the spacing uneven. In the most common design, the *transmon*, the $|1\rangle \to |2\rangle$ transition is a few percent lower in frequency than the qubit transition (Figure 1.2, right), so a pulse at f_{01} drives only the qubit.

Transmon frequencies are typically 4-6 GHz, in the microwave range. At room temperature, the thermal energy is about a thousand times larger than hf, so heat alone would excite the qubit at random. Superconducting chips therefore sit in a *dilution refrigerator* at about 10 mK, a hundredth of a degree above absolute zero (Exercise 1.1), and cables bring the microwave pulses down to them from electronics at room temperature.

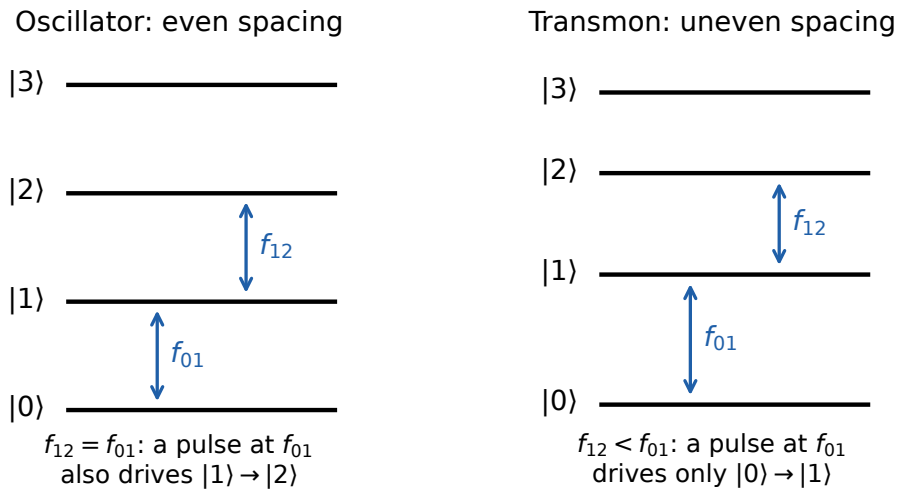
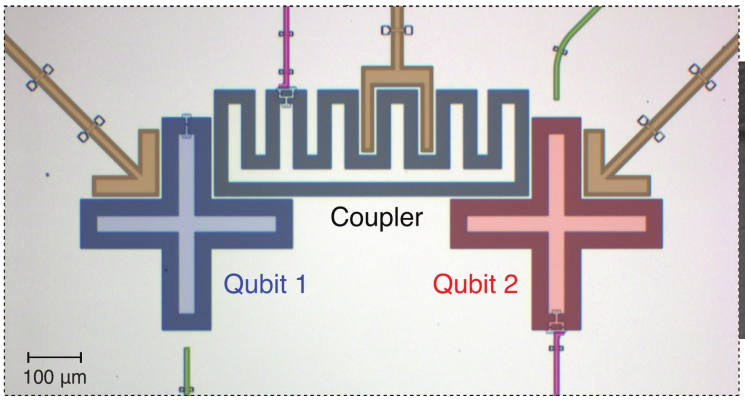


Figure 1.2: Energy levels of an oscillator and of a transmon. With even spacing, every transition has the same frequency, so the two lowest levels cannot be driven on their own. The transmon's uneven spacing makes the qubit transition unique. Not to scale: in a real transmon, f_{12} is only a few percent below f_{01}.

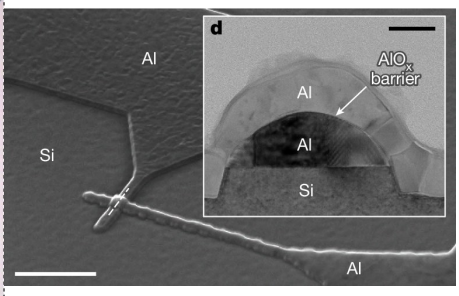
The cold also prepares the qubits. At 10 mK, a qubit left alone loses its energy and falls into $|0\rangle$ within a few hundred microseconds. This *relaxation* is a source of noise during a computation (Section 1.5), but between runs it initialises the qubits: we simply wait. A faster approach to initialisation is *active reset*: measure the qubit (Section 1.4.1) and apply an X gate if the result is 1.

Superconducting qubits are fast. A transmon is large for a quantum system, a fraction of a millimetre across, and it behaves like a small antenna that couples strongly to microwaves, so a pulse of a few tens of nanoseconds is enough for a single-qubit gate. The same antenna would pick up any stray radiation, so the chip is shielded, inside a metal package and the nested shields of the refrigerator, and its control lines are heavily filtered. Ideally, these lines are the qubit's only contact with the outside world. The shielding is never perfect, and the quantum state of a qubit typically survives for tens to hundreds of microseconds.

Each qubit has its own frequency, and this is how the qubits on a chip are addressed separately. Every qubit usually has its own control line, but part of a pulse leaks to the neighbouring qubits, and coupled qubits feel each other's drive. Neighbouring qubits are therefore given different frequencies, so that a pulse meant for one of them is off resonance for the others (Figure 1.1). In many designs, the frequency can also be tuned. The Josephson junction is then replaced by a small loop with two junctions, and a magnetic field through the loop, set by the current in a nearby control line, changes the frequency of the qubit. Tuning lets the electronics move qubits towards each other in frequency and away again, which is one way to switch on a two-qubit gate.



(a) Two transmons and a coupler.



(b) A Josephson junction.

Figure 1.3: Superconducting qubits under the microscope. (a) Two transmon qubits (blue and red, false colour) with a third transmon between them, used as a tunable coupler (grey) (Sung et al. 2021). The cross-shaped pads are the transmons' capacitors, and their Josephson junctions sit in the small structures at the ends of the arms. Qubit 2 and the coupler have two junctions in a loop, so that their frequencies can be tuned. Green lines carry the drive pulses, pink lines the currents that tune the frequencies, and the brown structures belong to the readout (Section 1.4.1). (b) A Josephson junction from a different device, seen in an electron microscope: two aluminium wires (Al) cross on a silicon chip (Si), separated by a thin oxide barrier, shown in cross-section in the inset (Van Damme et al. 2024). Scale bars: 1 µm; inset, 50 nm. Both images adapted (cropped) under the CC BY 4.0 licence.

Fabrication is never perfectly precise, so no two qubits come out identical. Each has its own frequency, its own coherence time and its own error rates. Every pulse must therefore be *calibrated*: its frequency, strength and duration are measured and adjusted, qubit by qubit and pair by pair, in a long sequence of automated experiments. The properties of the qubits also drift, over hours to days, so calibration is repeated regularly, and providers publish the resulting calibration data for each qubit (Section 1.5).

Two qubits can interact only if they are coupled, directly through a capacitor or through a small circuit between them, a *coupler* (Figure 1.3). The couplings are built into the chip, so which pairs can interact is fixed when the chip is made (Section 1.3.6). A coupling makes an interaction possible, and the electronics decide when it acts. Like a qubit and a pulse, two coupled qubits exchange energy efficiently only when their frequencies match. While their frequencies are kept apart, or the coupler is tuned to cancel the coupling, they hardly interact. A two-qubit gate switches the interaction on for a short time, by tuning the qubits or the coupler, or by driving one of the qubits (Section 1.3.2). The interaction never switches off completely, and what remains is a source of errors called *crosstalk*.

Exercise 1.1 (Thermal excitation)**.** In thermal equilibrium at temperature T, the excited and ground states of a qubit with frequency f are populated in the ratio $p_1/p_0 = e^{-hf/k_BT}$ (the Boltzmann factor), where k_B is Boltzmann's constant. Use $h/k_B \approx 0.048$ K/GHz.

(a) Compute p_1/p_0 for a 5 GHz qubit at room temperature (300 K), at 1 K and at 10 mK.

(b) Below which temperature is p_1/p_0 smaller than 10^{-3}?

Hint: Take the logarithm of both sides.

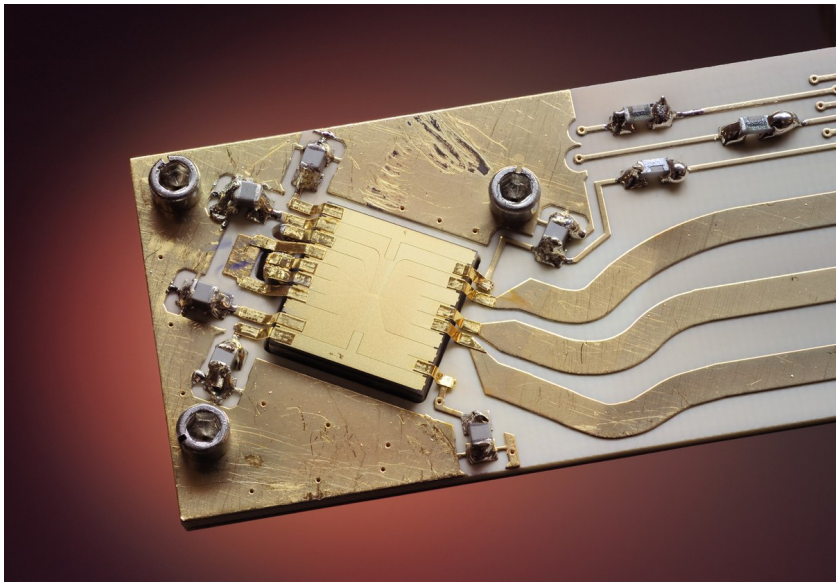
1.2.4 Trapped ions

A trapped-ion qubit is a single *ion*, an atom with one electron removed, typically of ytterbium, barium or calcium. The electrons of an atom can only be in certain states, each with its own energy. These are the atom's *internal* energy levels, as opposed to the motion of the atom as a whole, which will also matter below. Two of these levels form the qubit. All ions of an element are exactly the same, so, unlike fabricated circuits, the qubits of an ion-trap computer are identical by nature. As a consequence, they all have the same frequency, and a pulse cannot select one qubit by its frequency. Instead, most designs aim a laser beam at a single ion (Figure 1.5). In exchange, calibration is simpler, since all qubits share their frequency and most of their properties.

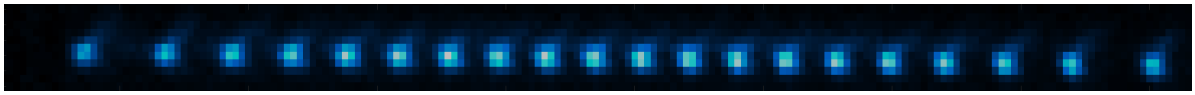
The ions are held in vacuum by electric fields in an *ion trap*, lined up in a chain a few micrometres apart, and laser beams slow them down until they are almost at rest (*laser cooling*). Another laser pulse then prepares each ion in $|0\rangle$ within microseconds (*optical pumping*). Laser beams, or in some designs microwaves, also drive the gates. The qubit levels can be chosen so that stray magnetic fields and other disturbances hardly affect them, and trapped-ion qubits keep their quantum state for a second or more, thousands of times longer than superconducting qubits. Figure 1.4 shows a trap chip and a chain of ions.

The ions are charged and repel each other, so the chain behaves like a row of balls joined by springs (Figure 1.5). Push one ion, and the whole chain starts to vibrate. Like the internal energy of an atom, the energy of these vibrations comes in discrete units, called *phonons*. Every ion takes part in the same vibrations, so they act as a shared channel, the *phonon bus*, through which any ion can interact with any other. The two-qubit gates are built by exploiting this phonon bus, so they can be applied to any pair of qubits: the connectivity is all-to-all (Section 1.3.6).

This comes at the cost of speed. Gates that use the phonon bus take from tens of microseconds to about a millisecond, roughly a thousand times longer than superconducting gates, and they become slower and harder to control as the chain grows. To scale up, some devices, such as Quantinuum's, divide the trap into zones and move ions between them, bringing the two ions of each gate together (*QCCD*).



(a) An ion-trap chip.



(b) A chain of twenty ions.

Figure 1.4: Trapped ions. (a) An ion trap made at NIST: a square gold chip, 7.4 mm on a side, on its circuit board. In experiments, two ions hover above the middle of the chip, and microwaves fed in through the three thick lines at the lower right drive their gates (photo: Y. Colombe/NIST, public domain). (b) Twenty calcium ions in a linear trap at Innsbruck, seen through the light they scatter: each bright spot is one ion, and the chain is about 0.1 mm long (Friis et al. 2018). Adapted (cropped) under the CC BY 4.0 licence.

1.2.5 Other platforms

Several other platforms are developing quickly, following the same recipe of a two-level system, a way to drive it, an interaction between qubits and a readout. In *neutral-atom* devices, uncharged atoms are held, and can be moved, by tightly focused laser beams (*optical tweezers*), and they interact when lasers excite them to very large *Rydberg states*; arrays of hundreds to thousands of atoms have been built (Henriet et al. 2020). *Photonic* devices carry qubits on particles of light, which hardly interact with their surroundings but, for the same reason, hardly with each other either, so their entangling operations succeed only some of the time (Slussarenko and Pryde 2019). *Spin qubits* use the *spin* of single electrons trapped in tiny semiconductor structures (*quantum dots*); they can be made with the techniques of the chip industry, but current devices have at most a few tens of qubits (Burkard et al. 2023). Table 1.2

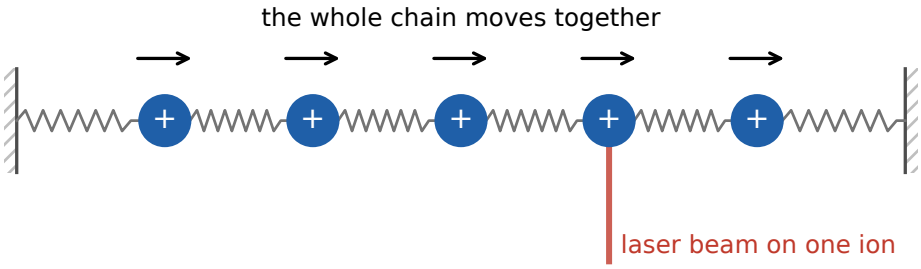


Figure 1.5: A trapped-ion chain as balls joined by springs. The ions repel each other, so they move together; shown here is the vibration in which the whole chain moves as one. A laser beam focused on a single ion drives that ion's qubit.

compares the five platforms.

Table 1.2: Platforms for gate-based quantum computing, compared.

Platform	Qubit	Two qubits interact through	Strengths	Main limitations
Superconducting	energy levels of a circuit on a chip	couplings built into the chip	fast gates; mature fabrication technology	short coherence; fixed connectivity; qubits not identical
Trapped ions	energy levels of an ion	shared vibrations of the chain	identical qubits; long coherence; any pair can interact	slow gates; long chains are hard to control
Neutral atoms	energy levels of an atom	excitation to large Rydberg states	many qubits; atoms can be rearranged	slow; atoms get lost and must be replaced
Photonics	a photon	interference and detection	little decoherence; photons travel far	entangling operations often fail; photons get lost
Spin qubits	spin of an electron	neighbouring electrons	very small; chip-industry fabrication	few qubits so far; qubits not identical

1.3 Native gates and connectivity

The circuits we write can use any gate that is convenient. A device, however, implements only a small set of *native gates*, each one a calibrated pulse or a short sequence of pulses, and it applies two-qubit gates only between pairs of qubits that are physically coupled. An additional step, compilation, translates every circuit into these native operations. This section explains which gates are native and why, and what the translation costs.

1.3.1 Pauli rotations

Recall the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They are Hermitian and square to the identity, $X^2 = Y^2 = Z^2 = I$. Any two different Pauli matrices anticommute, for example $XZ = -ZX$, and the product of two of them is the third, up to a factor: $XY = iZ$, $YZ = iX$ and $ZX = iY$. On several qubits, we combine them with tensor products.

Definition 1.1 (Pauli string)**.** A *Pauli string* on n qubits is a tensor product $P = P_1 \otimes P_2 \otimes \cdots \otimes P_n$ with each $P_j \in \{I, X, Y, Z\}$. We often drop the tensor product signs and write, for example, ZZ or XIY.

A Pauli string is again Hermitian, and it squares to the identity, because the square of a tensor product is the tensor product of the squares: $P^2 = P_1^2 \otimes \cdots \otimes P_n^2 = I$. This one property gives a closed formula for the exponential of P. Expand the exponential as a power series. Even powers of P are the identity, $P^{2m} = I$, and odd powers are $P^{2m+1} = P$, so the series splits into two:

$$e^{-i\theta P/2} = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} \left(\frac{\theta}{2}\right)^{2m} I \;-\; i \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} \left(\frac{\theta}{2}\right)^{2m+1} P.$$

These are the power series of the cosine and the sine, so

$$e^{-i\theta P/2} = \cos\frac{\theta}{2}\, I - i \sin\frac{\theta}{2}\, P. \tag{1.1}$$

The derivation used only $P^2 = I$, so Equation 1.1 holds for any operator with that property. Such operators are called *involutory*. For Hermitian operators, like all those in this chapter, being involutory is the same as being a *reflection*: the eigenvalues are ± 1, so the operator flips the sign of one subspace and leaves the orthogonal subspace unchanged.

Applied to single Pauli matrices, Equation 1.1 gives the familiar rotations of a qubit about the axes of the Bloch sphere, for example

$$R_X(\theta) = e^{-i\theta X/2} = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \qquad R_Z(\theta) = e^{-i\theta Z/2} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}. \tag{1.2}$$

Rotations about two different axes are enough to build any single-qubit gate. Up to a global phase, every single-qubit unitary can be written as $R_Z(\alpha)\,R_Y(\beta)\,R_Z(\gamma)$ for some angles (Nielsen and Chuang 2010, sec. 4.2).

Applied to the two-qubit Pauli string $Z \otimes Z$, Equation 1.1 gives

$$R_{ZZ}(\theta) = e^{-i\theta\, Z\otimes Z/2} = \begin{pmatrix} e^{-i\theta/2} & 0 & 0 & 0 \\ 0 & e^{i\theta/2} & 0 & 0 \\ 0 & 0 & e^{i\theta/2} & 0 \\ 0 & 0 & 0 & e^{-i\theta/2} \end{pmatrix} \tag{1.3}$$

in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. R_{ZZ} applies a phase that depends only on whether the two bits are equal or different, their parity. That is why it can be built from two CNOTs and a single R_Z (Figure 1.6). The first CNOT writes the parity of the two bits into the second qubit, R_Z applies the phase, and the second CNOT undoes the first. Section 1.3.4 extends this construction to rotations about any Pauli string, built from native gates.

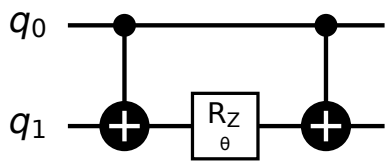


Figure 1.6: $R_{ZZ}(\theta)$ built from two CNOTs and one $R_Z(\theta)$.

Equation 1.1 will come back several times in this course, in circuits for time evolution, which are built from Pauli rotations, and in the parameter-shift rule, which gives exact gradients of variational circuits.

1.3.2 Physical implementation of gates

The native gates of a qubit platform are set by the physical mechanisms that control its qubits. This section describes these mechanisms, for single-qubit and for two-qubit gates, and the gates they produce.

A single-qubit gate on a superconducting qubit is a microwave pulse at the qubit's frequency, a few tens of nanoseconds long (Section 1.2.2). The area under the envelope of the pulse, set by its strength and duration, sets the rotation angle θ (Figure 1.1). The phase of the pulse, that is, the timing of the microwave oscillation relative to a reference clock in the control electronics, sets the axis of rotation, which lies in the xy-plane of the Bloch sphere at an angle φ from the x-axis. Delaying the oscillation by a fraction of its period turns the axis by the same fraction of a full turn. Laser or microwave pulses on trapped ions work in the same way. On both platforms, the natural single-qubit gate is therefore a rotation by any angle about any axis in the xy-plane, which IQM calls PRX:

$$\mathrm{PRX}(\theta, \varphi) = \exp\left[-i\frac{\theta}{2}\left(\cos\varphi\, X + \sin\varphi\, Y\right)\right]. \tag{1.4}$$

The operator $\cos\varphi\, X + \sin\varphi\, Y$ squares to the identity (Exercise 1.2), so Equation 1.1 applies to it too. For $\varphi = 0$ and $\varphi = \pi/2$, PRX reduces to $R_X(\theta)$ and $R_Y(\theta)$. Figure 1.7 shows how PRX rotates the Bloch sphere. Rotations about the z-axis are missing from this family, and Section 1.3.5 shows that they need no pulse at all.

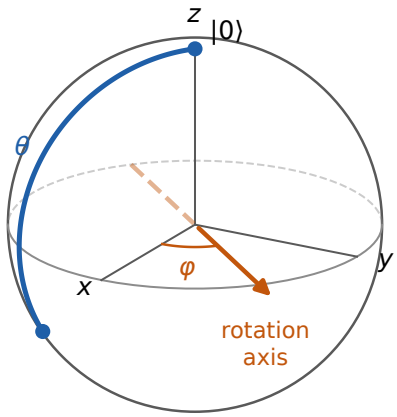


Figure 1.7: The Bloch sphere rotated by $\mathrm{PRX}(\theta, \varphi)$, a rotation by θ about the axis in the xy-plane at an angle φ from the x-axis. The blue path shows where $|0\rangle$ goes as the rotation angle grows from 0 to θ. In the web version, the figure is interactive: the sliders set θ and φ, the blue grid turns with the sphere, and dragging changes the view. Below the sphere, a schematic of the pulse: its area, set by its amplitude at a fixed duration, gives θ, and its phase, a delay against the reference clock (dashed), gives φ. The button plays the pulse, and the sphere turns as the area builds up.

Exercise 1.2 (The PRX gate)**.**

(a) Show that $(\cos\varphi\, X + \sin\varphi\, Y)^2 = I$.

Hint: Expand the square, and use that X and Y anticommute.

(b) Write $\mathrm{PRX}(\theta, \varphi)$ as a 2×2 matrix.

Hint: Use Equation 1.1, which applies because of (a).

(c) Check that $\mathrm{PRX}(\theta, 0) = R_X(\theta)$ and $\mathrm{PRX}(\theta, \pi/2) = R_Y(\theta)$. Which rotation is $\mathrm{PRX}(\theta, \pi)$?

During a two-qubit gate, the two qubits should interact, and at all other times they should not (Section 1.2.3). Superconducting devices switch the interaction on in three main ways, and each gives a different native gate.

- **Tuning two qubits into resonance.** When two coupled qubits have the same frequency, an excitation moves back and forth between them: $|01\rangle$ turns into $|10\rangle$ and back again, while $|00\rangle$ and $|11\rangle$ are unchanged. A complete transfer gives the iSWAP gate, which exchanges $|01\rangle$ and $|10\rangle$ and multiplies both by i. Stopping halfway gives its square root, $\sqrt{\mathrm{iSWAP}}$.
- **Using the third level.** A transmon has a third level, $|2\rangle$, not far above the qubit levels (Figure 1.2). The frequencies can be tuned so that the two-qubit state $|11\rangle$ has the same energy as $|20\rangle$, in which the first transmon holds both excitations. These two states then exchange instead. After one complete exchange, there and back, $|11\rangle$ returns with its sign flipped (up to phases on the single qubits, which are easy to correct), while $|00\rangle$, $|01\rangle$ and $|10\rangle$, which have no partner at the same energy, are unchanged. This is the controlled-Z gate, CZ, which flips the sign of $|11\rangle$ only. On most current devices, a tunable coupler between the qubits switches the interaction, and CZ is the most common native gate.
- **Driving one qubit at its neighbour's frequency.** IBM's older Eagle devices, retired in 2026, have qubits with fixed frequencies and couplings that are always on. A pulse on one qubit at the frequency of its neighbour (*cross-resonance*) makes the neighbour rotate about its x-axis, in a direction that depends on the state of the first qubit: a rotation generated by $Z \otimes X$. In practice, it is applied as a gate called ECR, for echoed cross-resonance, which equals a CNOT up to single-qubit gates.

On trapped ions, laser beams on two ions couple their qubit states to the shared vibrations of the chain, the phonon bus of Section 1.2.4. The light pushes each ion with a force that depends on its qubit state, so the chain vibrates in a way that depends on both qubits, and through this vibration the two qubits interact. At the end of the gate the vibration returns to its starting state, so that it carries away no information about the qubits. This *Mølmer-Sørensen gate* is a rotation generated by $X \otimes X$, of the form of Equation 1.1, with an angle set by the duration and strength of the laser pulses. It is available at any angle, from small rotations up to a maximally entangling one, and between any pair of ions in the chain. Combined

with single-qubit rotations, it gives $R_{ZZ}(\theta)$ at any angle, the native two-qubit gate of current trapped-ion devices.

1.3.3 Native gate sets

Table 1.3 lists the native gates of some current devices. Names and details change with each hardware generation, but the pattern stays the same: a continuous family of single-qubit rotations, and one or a few entangling gates. Together they are universal, so any circuit can be compiled into them, at the cost of extra gates.

Table 1.3: Native gates of some current devices, as of 2026. All of them also offer virtual Z rotations (Section 1.3.5).

Device family	Single-qubit	Two-qubit
IQM (superconducting)	PRX(θ, φ)	CZ; on star devices, also MOVE to and from the central resonator
IBM (superconducting)	$\sqrt{X}$, X; experimental: $R_X(\theta)$	CZ (Heron, Nighthawk); ECR on the retired Eagle devices; experimental: $R_{ZZ}(\theta)$
Google (superconducting)	PhasedXZ	CZ (Willow); $\sqrt{\text{iSWAP}}$ and others on the older Sycamore
Quantinuum (trapped ions)	$R_{xy}(\theta, \varphi)$	$R_{ZZ}(\theta)$, any angle
IonQ (trapped ions)	GPi(φ), GPi2(φ)	MS (Aria), ZZ(θ) (Forte), any angle

The single-qubit gates in the table are rotations about axes in the xy-plane under different names, in one case followed by a Z rotation:

- $\sqrt{X}$ and X (IBM) are $R_X(\pi/2)$ and $R_X(\pi)$, up to a global phase; IBM's Heron devices also offer $R_X(\theta)$ at any angle, as an experimental option;
- PhasedXZ (Google) is an xy-plane rotation followed by a Z rotation;
- $R_{xy}(\theta, \varphi)$ (Quantinuum, also called U1q) is PRX(θ, φ) under another name;
- GPi(φ) and GPi2(φ) (IonQ) are PRX(π, φ) and PRX$(\pi/2, \varphi)$, up to a global phase. IonQ measures angles in *turns*, so that an angle of 1 is a full turn, 2π.

The two-qubit gates are either fixed gates or rotations with an adjustable angle. The fixed ones are CZ, $\sqrt{\text{iSWAP}}$ and ECR. In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$\text{CZ} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \qquad \sqrt{\text{iSWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

and ECR equals a CNOT up to single-qubit gates on both qubits. The adjustable ones are $R_{ZZ}(\theta)$ of Equation 1.3, IonQ's ZZ(θ), which is the same gate with its angle in turns, and the Mølmer-Sørensen gate MS, generated by $X \otimes X$. With an adjustable angle, an entangling rotation of any strength is a single native gate, which saves gates in circuits built from Pauli rotations.

1.3.4 Compiling into native gates

Compiling a circuit into native gates relies on an identity that we will use several times: conjugating an exponential conjugates its exponent,

$$U e^{A} U^{\dagger} = e^{UAU^{\dagger}} \tag{1.5}$$

for any unitary U and any square matrix A. To see this, expand both sides as power series: the inner factors $U^{\dagger}U = I$ cancel in $(UAU^{\dagger})^{k} = UA^{k}U^{\dagger}$. For a Pauli rotation, Equation 1.5 says that $Ue^{-i\theta P/2}U^{\dagger} = e^{-i\theta UPU^{\dagger}/2}$: conjugated by U, a rotation about P becomes a rotation about $UPU^{\dagger}$, by the same angle. To rotate about a long Pauli string, it is therefore enough to find a circuit U of native gates that turns a simple Pauli matrix, such as X on a single qubit, into that string.

Two kinds of conjugation do the job. First, single-qubit rotations turn one Pauli matrix into another; for example, $R_Y(\pi/2)$ turns Z into X, and $R_X(-\pi/2)$ turns Z into Y (Exercise 1.3). Second, CZ attaches a Z to an X on the other qubit:

$$\mathrm{CZ}\,(I \otimes X)\,\mathrm{CZ} = Z \otimes X.$$

To check this, apply both sides to a basis state $|ab\rangle$. On the left, the first CZ gives a sign $(-1)^{ab}$, X flips b, and the second CZ gives a sign $(-1)^{a(1-b)}$, so the result is $(-1)^{a}\,|a, 1-b\rangle$, which is the action of $Z \otimes X$. Each further CZ on the same qubit attaches one more Z: writing CZ_{jk} for a CZ on qubits j and k, and labelling single-qubit Paulis by their qubit, conjugation by $\mathrm{CZ}_{13}\mathrm{CZ}_{23}$ turns X_3 into $Z_1 Z_2 X_3$. Single-qubit conjugations then turn each factor into the Pauli matrix we need. Figure 1.8 shows the result for $P = Z \otimes Y \otimes X$. Every gate in it is native on a device with PRX and CZ, since $R_X(\alpha) = \mathrm{PRX}(\alpha, 0)$.

A rotation about a Pauli string on k qubits costs $2(k-1)$ CZs this way, the same number of two-qubit gates as the CNOT construction of Figure 1.6. In Figure 1.8, all CZs act on the qubit that carries the rotation, which suits a device where that qubit is coupled to all the others. On a line of qubits, a chain of CNOTs between neighbours collects the parity step by step instead, and each CNOT is one CZ between two single-qubit rotations (Exercise 1.6).

Exercise 1.3 (Turning one Pauli into another)**.** Let P and Q be Pauli strings that anticommute, $PQ = -QP$.

(a) Show that $e^{-i\alpha P/2}\, Q\, e^{i\alpha P/2} = \cos\alpha\, Q - i \sin\alpha\, PQ$.

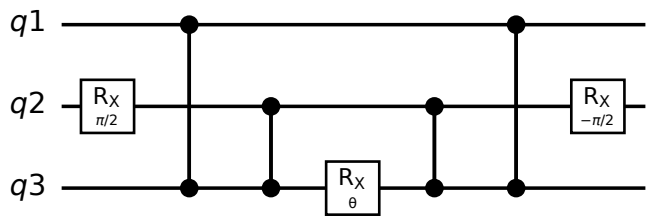


Figure 1.8: $e^{-i\theta\, Z\otimes Y\otimes X/2}$ built from CZ and R_X rotations. The CZs turn the X rotation on q_3 into a rotation about $Z\otimes Z\otimes X$, and the outer R_X rotations on q_2 turn its Z into Y.

Hint: Write both exponentials with Equation 1.1, and use $PQP = -Q$.

(b) Show that $R_Y(\pi/2)$ turns Z into X, and that $R_X(-\pi/2)$ turns Z into Y.

Hint: Use (a), and the products of Pauli matrices.

(c) Check that the circuit of Figure 1.8 implements $e^{-i\theta\, Z\otimes Y\otimes X/2}$.

Hint: Use Equation 1.5, the identity for CZ in the text, and (b).

(d) Draw a circuit of CZ gates and R_X or R_Y rotations for $R_{ZZ}(\theta)$, and compare it with Figure 1.6.

Hint: Start from $R_X(\theta)$ on the second qubit, conjugate it by CZ, then turn the remaining X into Z.

1.3.5 Virtual Z gates

PRX contains no rotations about the z-axis, yet circuits are full of them, for example in the decomposition $R_Z(\alpha)\, R_Y(\beta)\, R_Z(\gamma)$ of a general single-qubit gate (Section 1.3.1) and in many parametrised gates of algorithms. On most devices, superconducting and trapped-ion alike, Z rotations are implemented without any pulse, as *virtual Z gates*. The trick follows from Pauli algebra.

Conjugation by a Z rotation rotates X and Y within the xy-plane:

$$R_Z(\alpha)\, X\, R_Z(\alpha)^\dagger = \cos\alpha\, X + \sin\alpha\, Y, \qquad R_Z(\alpha)\, Y R_Z(\alpha)^\dagger = \cos\alpha\, Y - \sin\alpha\, X$$

(Exercise 1.4). By the angle-addition formulas, it therefore maps $\cos\varphi\, X + \sin\varphi\, Y$ to $\cos(\varphi+\alpha)\, X + \sin(\varphi+\alpha)\, Y$. By Equation 1.5, conjugating a PRX pulse by a Z rotation therefore shifts its phase:

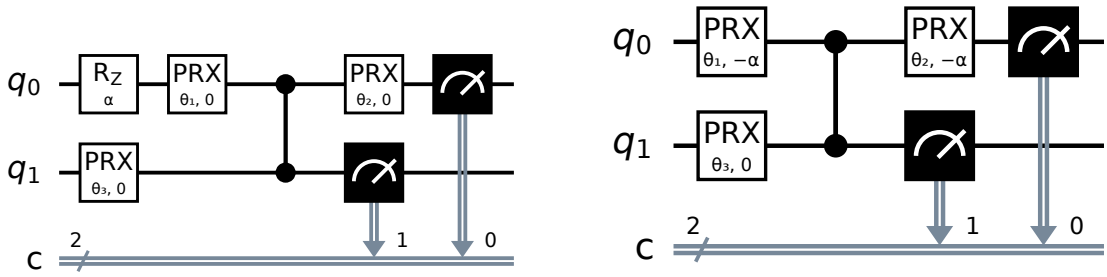
$$R_Z(\alpha)\, \mathrm{PRX}(\theta,\varphi)\, R_Z(\alpha)^\dagger = \mathrm{PRX}(\theta,\varphi+\alpha). \tag{1.6}$$

Replacing α by $-\alpha$ and multiplying by $R_Z(\alpha)$ on the left gives

$$\mathrm{PRX}(\theta, \varphi)\, R_Z(\alpha) = R_Z(\alpha)\, \mathrm{PRX}(\theta, \varphi - \alpha). \tag{1.7}$$

Operators act from right to left, so on the left-hand side $R_Z(\alpha)$ acts first and the pulse second. On the right-hand side, the pulse acts first, with its phase shifted by $-\alpha$. A Z rotation can therefore be moved later in a circuit, past a pulse, by changing the pulse's phase.

Now take a circuit with a Z rotation on one qubit (Figure 1.9a), and move the Z rotation forward in time. Each PRX it passes gets its phase shifted by $-\alpha$. CZ is diagonal, like R_Z, so the two commute and the Z rotation passes through unchanged. At the end, the Z rotation sits just before a measurement in the computational basis. A measurement only asks whether each qubit is in $|0\rangle$ or in $|1\rangle$ (Section 1.2.1), and the Z rotation only changes the phases of $|0\rangle$ and $|1\rangle$, not the probabilities of measuring them, so it can be dropped. The result (Figure 1.9b) contains no Z rotation at all.



(a) With a Z rotation on q_0.

(b) The Z rotation absorbed into the phases of the later pulses on q_0.

Figure 1.9: Two circuits that give the same measurement statistics.

In hardware, the phase of each pulse, microwave or laser, is measured against a reference clock in the control electronics, and a virtual Z gate shifts that reference for all later pulses on the qubit. It is an instruction to the electronics, carried out instantly and exactly, so Z rotations take no time and add no error. When we estimate the error of a circuit, only the other gates count.

The same trick makes every single-qubit gate cost a single pulse. Write the gate as $R_Z(\alpha)\, R_Y(\beta)\, R_Z(\gamma)$, note that $R_Y(\beta) = \mathrm{PRX}(\beta, \pi/2)$, and move $R_Z(\gamma)$ past the pulse with Equation 1.7:

$$R_Z(\alpha)\, R_Y(\beta)\, R_Z(\gamma) = R_Z(\alpha + \gamma)\, \mathrm{PRX}(\beta, \pi/2 - \gamma).$$

That is one pulse, followed by a Z rotation that is virtual.

Exercise 1.4 (Virtual Z gates)**.**

(a) Show that $R_Z(\alpha)\, X\, R_Z(\alpha)^\dagger = \cos\alpha\, X + \sin\alpha\, Y$ and $R_Z(\alpha)\, Y R_Z(\alpha)^\dagger = \cos\alpha\, Y - \sin\alpha\, X$.

Hint: Use Exercise 1.3 (a) with $P = Z$.

(b) Write the Hadamard gate, up to a global phase, as a single PRX pulse followed by a Z rotation.

Hint: Write the Hadamard gate as $R_Z(\alpha)\, R_Y(\beta)\, R_Z(\gamma)$ first, then use Equation 1.7.

1.3.6 Connectivity

A native two-qubit gate is available only between qubits that are physically coupled. The *coupling graph* of a device records which pairs these are. Its vertices are the qubits, and an edge joins two qubits if a native two-qubit gate can act on them (Figure 1.10).

On a superconducting chip, the coupling graph is fixed when the chip is made. In a square lattice, used by Google, IQM and IBM's newest devices, each qubit has up to four neighbours. IBM's heavy-hex lattice, used on most of its current devices, gives each qubit only two or three, which reduces unwanted interactions between qubits but lengthens the routes between distant ones. Some IQM devices use a star layout, in which every qubit couples to a central resonator, a circuit element that can take over the state of one qubit and pass it on to another; in Figure 1.10 it is drawn as an extra vertex.

In a trapped-ion chain, every ion couples to the same phonon bus, so every pair can interact and the coupling graph is complete. Devices that move ions between zones achieve the same by bringing any two ions together.

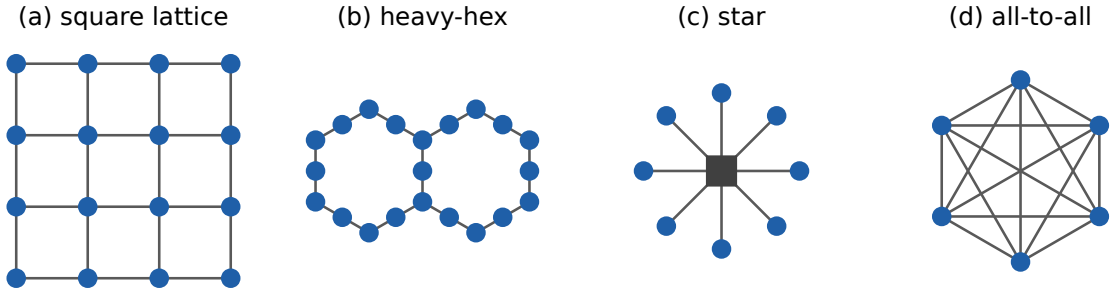


Figure 1.10: Coupling graphs: (a) square lattice; (b) heavy-hex lattice; (c) star, with every qubit coupled to a central resonator (square); (d) all-to-all, as in a trapped-ion chain.

What if a circuit needs a two-qubit gate between qubits that are not neighbours? Their states must first be brought together with *SWAP* gates, which exchange the states of two neighbouring qubits, and each SWAP costs three CNOTs (Figure 1.11). A gate between distant qubits can therefore cost many extra two-qubit gates (Exercise 1.5). Two-qubit gates are the main source of error on current devices (Section 1.5), so connectivity directly limits which circuits a device can run reliably.

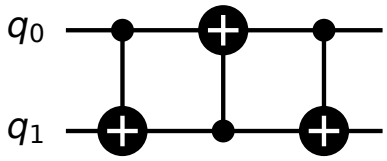


Figure 1.11: SWAP as three CNOTs with alternating control.

In practice, nobody inserts SWAPs by hand. *Transpilation* maps a circuit onto the native gates and coupling graph of a device. It chooses which physical qubit plays the role of each qubit in the circuit, inserts SWAPs where needed, and simplifies the result. We return to its cost in later chapters.

Exercise 1.5 (The cost of distance)**.**

(a) Check Figure 1.11 on the four computational basis states.

(b) We want to implement a CNOT between the qubits at opposite ends of a line of n qubits, using SWAPs. If the qubits may stay in their new positions after the operation, how many CNOTs are needed? And if they must be returned to their original positions?

Hint: Each SWAP moves a state one step along the line.

(c) If the qubits must be returned to their original positions, can you implement the same CNOT with fewer CNOTs, without SWAPs?

Hint: Start with $n = 3$, and follow what each CNOT does to the bits of a basis state $|abc\rangle$.

Exercise 1.6 (SWAP from native gates)**.** On a device whose native gates are PRX and CZ, a SWAP must be built from these.

(a) Let A be a single-qubit unitary. Show that $(I \otimes A)\,\mathrm{CZ}\,(I \otimes A^\dagger)$ applies $AZA^\dagger$ to the second qubit if the first is in $|1\rangle$, and does nothing if it is in $|0\rangle$.

Hint: Write $\mathrm{CZ} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes Z$.

(b) Find a PRX rotation A with $AZA^\dagger = X$, and write a CNOT as one CZ and two PRX gates.

Hint: Exercise 1.3 (b) gives a suitable A.

(c) Using Figure 1.11, write a SWAP with three CZs and PRX gates only. How many PRX pulses does it need?

1.4 Measurement

1.4.1 Readout

On superconducting devices, each qubit is coupled to a small resonant circuit, its readout resonator, whose frequency shifts slightly depending on whether the qubit is in $|0\rangle$ or $|1\rangle$. To measure, the electronics send a weak microwave pulse to the resonator and decide between 0 and 1 from the signal that comes back (*dispersive readout*). This takes about a microsecond. On trapped ions, a laser beam is tuned so that an ion in one qubit state scatters light and appears bright, while an ion in the other state stays dark. A camera or photon detectors count the light from each ion, and this fluorescence readout takes from a hundred microseconds to a millisecond.

On both platforms, a measurement returns one bit per qubit, in the computational basis (Section 1.2.1). It projects each qubit onto $|0\rangle$ or $|1\rangle$, and measuring in another basis means first rotating that basis onto the computational basis.

1.4.2 Shots and expectation values

A single run of a circuit, from preparing the qubits to measuring them, is called a *shot*. It returns one bit string, drawn at random from the distribution given by the Born rule, so to learn anything quantitative we repeat the circuit many times.

Most quantities in this course are expectation values $\langle P\rangle$ of Pauli strings. To estimate one, we rotate each qubit so that its factor of P becomes Z (nothing for Z, a Hadamard for X, and $R_X(\pi/2)$ for Y), and measure. A Z measurement with outcome bit b corresponds to the eigenvalue $(-1)^b$ of Z, so the outcome of P is the product of these signs over the qubits on which P acts:

$$s = (-1)^{\sum_{j:\,P_j \neq I} b_j} = \pm 1.$$

For $P = ZZ$, for example, $s = +1$ if the two bits agree and $s = -1$ if they differ. The average of s over N shots,

$$\bar{s} = \frac{1}{N} \sum_{k=1}^{N} s_k, \tag{1.8}$$

is our estimate of $\langle P \rangle$. A rule like this, which turns data into an estimate, is called an *estimator*.

1.4.3 Shot noise

How many shots does an estimate need? Each shot gives $s = +1$ or $s = -1$ at random, and the expectation value of s is $\langle P \rangle$. The sample mean $\bar{s}$ of Equation 1.8 is therefore correct on average, or *unbiased*. Its random error comes from the spread of single shots. Since $s^2 = 1$ for every shot (this is $P^2 = I$ again), the variance of a single shot is

$$\mathrm{Var}(s) = \langle s^2 \rangle - \langle s \rangle^2 = 1 - \langle P \rangle^2.$$

The mean of N independent samples has $1/N$ times the variance of a single sample (for a refresher on this and other basic statistics, see Blitzstein and Hwang (2019)). The typical size of the random error of $\bar{s}$, its *standard error*, is therefore

$$\sigma_{\bar{s}} = \sqrt{\frac{1 - \langle P \rangle^2}{N}}. \tag{1.9}$$

In practice $\langle P \rangle$ is unknown, and we evaluate Equation 1.9 with $\bar{s}$ in its place.

The error is largest when $\langle P \rangle = 0$, where every shot is a coin flip, and vanishes when $\langle P \rangle = \pm 1$, where every shot gives the same result. It decreases only as $1/\sqrt{N}$, so each extra digit of precision costs a hundred times more shots. To estimate an expectation value near zero with a standard error of 0.01, for example, we need $N = 1/0.01^2 = 10^4$ shots. Section 1.6 shows how long they take on each platform.

Shot noise is one of two kinds of error in an estimated expectation value. Noise in the device (Section 1.5) also shifts $\langle P \rangle$ itself, usually towards zero, and more shots cannot remove that shift; they only estimate the shifted value more precisely. The next chapter separates these two kinds of error (Section 2.7). Tutorial 1 practises Equation 1.8 and Equation 1.9 in code: open the T1 notebook in Colab.

1.5 Sources of noise

Every wire or laser beam used to control a qubit is also a way in for disturbances. This section describes in words what can go wrong; Chapter 2 goes deeper into the mathematical description of noise and its effects.

The most important effect is that an excited qubit loses its energy. Left alone, $|1\rangle$ decays to $|0\rangle$. This is the relaxation that initialises superconducting qubits between runs (Section 1.2.3), but during a computation it corrupts the state. In superconducting qubits, the energy is absorbed by tiny defects in the materials of the chip. The qubit levels of trapped ions hardly decay on their own, but the lasers used for gates occasionally knock an ion out of its qubit state.

A qubit can also lose its phase without losing energy. A superposition of $|0\rangle$ and $|1\rangle$ carries a relative phase between its two parts, and the qubit's frequency sets how fast this phase turns. If the frequency fluctuates, for example because of fluctuating magnetic fields or electric charges near the qubit, the phase wanders, and the superposition gradually becomes a random mixture of $|0\rangle$ and $|1\rangle$. This is called *dephasing*. The time over which a qubit relaxes is called T_1, and the time over which it dephases is called T_2. These are the coherence times quoted for every device, and the next chapter defines them precisely (Definition 2.5 and Definition 2.6).

A third kind of error is a gate that rotates by the wrong amount, because a pulse is slightly too long, too short or off axis, or because a laser's intensity fluctuates. Unlike relaxation and dephasing, such *coherent errors* are systematic: the same gate makes the same mistake every time. Table 1.4 lists these effects together with the remaining ones.

Table 1.4: What can go wrong in a quantum device.

What happens	Example	Name in the next chapter
$\|1\rangle$ decays to $\|0\rangle$	energy absorbed by defects in the chip	relaxation (T_1, Section 2.2.5)
the relative phase of a superposition wanders	fluctuating magnetic fields	dephasing (T_2, Section 2.2.5)
a gate rotates by the wrong amount	a pulse slightly too long	coherent error (Section 2.4)
the qubit leaves the qubit levels	a pulse pushes a transmon into its third level (Figure 1.2)	*leakage* (Section 2.4)
an operation on one qubit disturbs another	couplings that are never fully off; a laser beam that partly hits the next ion	crosstalk (Section 2.3)
the readout is wrong	0 reported as 1, or the reverse	readout error (Section 2.5)
many qubits fail at once	a cosmic ray deposits energy in the chip and disturbs all qubits nearby	correlated errors (Section 2.3)
the device drifts	properties that change over hours	(recalibration)

The last row explains why providers recalibrate their devices regularly and publish calibration data, the error rates of each qubit and gate at the last calibration. These data are a snapshot,

and a few hours later the device may behave differently.

Providers summarise the quality of each operation by a *fidelity*. For now, read it as one minus the probability that the operation goes wrong; the next chapter gives the precise definition (Definition 2.9). Typical values on current devices are 99.9-99.99% for single-qubit gates, 99-99.9% for two-qubit gates and 96-99.9% for readout, so two-qubit gates and readout are the weakest links.

1.6 Timescales

Table 1.5 collects typical timescales of current devices. They vary between devices and improve every year, so read them as orders of magnitude.

Table 1.5: Typical timescales of current devices, as orders of magnitude.

Time per	Superconducting	Trapped ions
single-qubit gate	20-60 ns	10-150 µs
two-qubit gate	20-700 ns	50 µs-1 ms
readout	0.5-2.5 µs	0.1-1 ms
coherence (T_2)	20-350 µs	about 1 s or more
shot (reset, circuit, readout)	0.3-4 ms	10 ms to several seconds
switching to a new circuit	much longer than a shot, plus any queue	the same

The coherence time divided by the two-qubit gate time is between about a thousand and ten thousand on both platforms, although their speeds differ by a factor of about a thousand. Both platforms thus meet DiVincenzo's requirement of coherence times that are long compared with gate times (Section 1.2.1), by a similar margin and in opposite ways: superconducting qubits with fast gates, trapped ions with long coherence. Coherence alone would allow circuits of thousands of gates in sequence. In practice, gate errors of 0.1-1% per two-qubit gate limit useful circuits much earlier.

Where the time goes differs between the platforms. On superconducting devices, the gates are fast, and most of a shot is spent measuring the qubits and resetting them to $|0\rangle$; devices run hundreds to thousands of shots per second. Switching to a different circuit costs much more, since the circuit must be compiled and loaded into the control electronics, and on a shared cloud device it may wait in a queue. Many shots of one circuit are cheap; many different circuits are expensive. On trapped ions, the gates themselves are slow, and each shot also includes cooling the ions and, on some devices, moving them between zones, which takes most of the time. There, the number of shots is the main cost. For the 10^4 shots of the example in

Section 1.4.3, this means between a few seconds and a minute on a superconducting device, and from a few minutes to several hours on an ion trap.

Exercise 1.7 (A shot budget)**.**

(a) How many shots give a standard error of 0.01 when $\langle P \rangle \approx 0.9$? Compare with $\langle P \rangle \approx 0$.

Hint: Solve Equation 1.9 for N.

(b) You need ten expectation values, each from its own circuit, each with a standard error of at most 0.005. Assume the worst case, $\langle P \rangle = 0$, and assume that loading each circuit takes 10 s, with no queue. Estimate the total time on a superconducting device and on an ion trap. Which part of the time dominates on each?

Hint: Use the times per shot of Table 1.5.

1.7 More exercises

These exercises combine material from several sections of the chapter.

Exercise 1.8 (An error budget)**.** A device has the native gates PRX and CZ, with virtual Z rotations. Its operations fail independently, with probability 10^{-4} for a PRX pulse, 5×10^{-3} for a CZ and 10^{-2} for a measurement. On a line of 10 qubits, numbered 0 to 9, we run a brick-wall circuit: each layer applies R_Z and R_X to every qubit, followed by CZs between neighbouring qubits, on the pairs (0,1), (2,3), … in the first layer and every second layer after it, and on the pairs (1,2), (3,4), … in the other layers. Figure 1.12 shows two layers on four qubits. The circuit has 8 layers and ends by measuring all qubits.

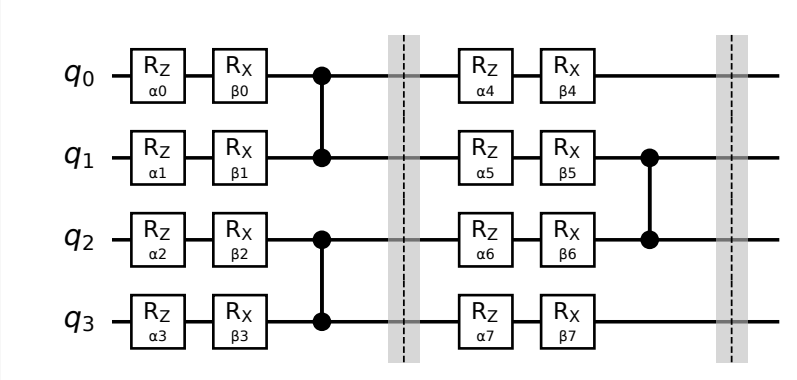


Figure 1.12: Two layers of a brick-wall circuit on four qubits.

(a) Count the native operations of each kind, and estimate the probability that a run of the circuit has no error at all. Which kind of operation contributes most to the failures?

Hint: Which gates in Figure 1.12 need a pulse?

(b) How does the answer to (a) change on a device without virtual Z rotations, where each R_Z needs a pulse with the same error as PRX?

(c) Roughly how many layers can the circuit have before the probability of a run without error drops below one half?

Hint: $(1-p)^n \approx e^{-np}$ for small p.

Exercise 1.9 (Readout errors bias the estimate)**.** Suppose that the readout of each qubit reports the wrong bit with probability p, independently in every shot and for both outcomes.

(a) Show that the recorded sign s of a single-qubit Z measurement has expectation value $(1-2p)\langle Z\rangle$.

(b) For a Pauli string P acting on k qubits, show that the recorded sign has expectation value $(1-2p)^k\langle P\rangle$.

Hint: The recorded sign is the true sign times one independent factor ± 1 per qubit.

(c) Take $p = 0.02$, $k = 4$ and $\langle P\rangle = 0.5$. Compare the bias of $\bar{s}$ with its standard error for $N = 10^3$ and for $N = 10^5$ shots. When do more shots stop helping?

Hint: The standard error is given by Equation 1.9.

(d) If p is known, how can the estimate be corrected? What does the correction do to the standard error?

Exercise 1.10 (Measuring in other bases)**.** A device has the native gates PRX and CZ, and measures in the computational basis.

(a) Which PRX pulses must be applied before the measurement to estimate $\langle X\otimes Y\otimes Z\rangle$?

Hint: On each qubit, you need a rotation U with $UP_jU^\dagger = Z$; see Exercise 1.3.

(b) Can $\langle X\otimes I\rangle$ and $\langle X\otimes X\rangle$ be estimated from the same shots? And $\langle X\otimes X\rangle$ and $\langle Z\otimes Z\rangle$, if only single-qubit rotations are allowed before the measurement?

1.8 Further reading

For superconducting qubits, Krantz et al. (2019) is a thorough guide written for engineers; for trapped ions, see Bruzewicz et al. (2019). Henriet et al. (2020), Slussarenko and Pryde (2019) and Burkard et al. (2023) review neutral atoms, photonics and spin qubits. Preskill (2018) introduced the NISQ era, and Nielsen and Chuang (2010, chap. 4) covers the circuit model.

2 Noise channels and error budgets

In the previous chapter, we described in words what can go wrong in a quantum device, its noise (Section 1.5). In this chapter, we describe noise mathematically. The goal is to model how noise affects the results of a computation, and to use these models in our estimates and simulations. One model runs through the whole chapter: a noisy operation is the ideal operation followed by a *quantum channel*, a map that turns the state of the qubits into a noisier one.

By the end of the chapter, you should be able to write a channel with Kraus operators and work out what it does to Pauli expectation values; relate T_1, T_2^* and T_2 to the channels behind them; tell local from global noise, and coherent from incoherent errors; model readout errors; estimate how much signal a circuit keeps; and separate the bias caused by noise from shot noise.

2.1 Density matrices and channels

We first review the two tools that the rest of the chapter uses throughout. If you know them well, skim this section for the notation.

2.1.1 Mixed states

Noise makes the state of a device uncertain. If the qubits are in the state $|\psi_k\rangle$ with probability p_k, the expectation value of an observable O is

$$\sum_k p_k \langle\psi_k| O |\psi_k\rangle = \mathrm{Tr}[O\rho], \qquad \text{with} \quad \rho = \sum_k p_k |\psi_k\rangle \langle\psi_k| .$$

The probability of each measurement outcome is the expectation value of a projector, so all predictions depend on the ensemble only through ρ.

Definition 2.1 (Density matrix)**.** A *density matrix* on n qubits is a $2^n \times 2^n$ matrix ρ that is positive semidefinite (Hermitian, with no negative eigenvalues) and has trace one. The expectation value of an observable O is $\langle O\rangle = \mathrm{Tr}[O\rho]$. A pure state $|\psi\rangle$ has $\rho = |\psi\rangle\langle\psi|$, and then $\langle O\rangle = \langle\psi| O |\psi\rangle$. Every other state is *mixed*.

A density matrix is pure exactly when $\rho^2 = \rho$, or, equivalently, when $\mathrm{Tr}[\rho^2] = 1$. The *purity* $\mathrm{Tr}[\rho^2]$ measures how mixed a state is. The eigenvalues λ_k of ρ are non-negative and sum to one, and the purity is $\sum_k \lambda_k^2$. It is 1 when one eigenvalue is 1 and the others are 0, which is a pure state. It is smallest, 2^{-n}, when all 2^n eigenvalues are equal, which is the *maximally mixed state* $I/2^n$: the qubits are in a uniformly random basis state, and we know nothing about them.

For a single qubit, we want to extend the Bloch-sphere picture of pure states (Figure 1.7) to mixed states. Every Hermitian 2×2 matrix can be written in the Pauli basis I, X, Y, Z, with real coefficients. The Pauli matrices have trace zero, and the product of two different ones is the third, up to a factor (Section 1.3.1), so $\mathrm{Tr}[PQ] = 2$ if $P = Q$ and 0 otherwise. If we write $\rho = \frac{1}{2}(c_I I + c_x X + c_y Y + c_z Z)$, the trace of $P\rho$ therefore picks out one coefficient at a time: $c_I = \mathrm{Tr}[\rho] = 1$ and $c_x = \mathrm{Tr}[X\rho] = \langle X \rangle$, and likewise for y and z. A density matrix of one qubit is therefore

$$\rho = \frac{1}{2}\left(I + r_x X + r_y Y + r_z Z\right), \qquad r_x = \langle X \rangle,\ r_y = \langle Y \rangle,\ r_z = \langle Z \rangle. \tag{2.1}$$

The vector $\vec{r} = (r_x, r_y, r_z)$ is the *Bloch vector*. The matrix $r_x X + r_y Y + r_z Z$ squares to $|\vec{r}|^2 I$, since the Pauli matrices anticommute. The eigenvalues of ρ are therefore $(1 \pm |\vec{r}|)/2$, and ρ is a density matrix exactly when $|\vec{r}| \leq 1$. Pure states lie on the surface of the Bloch sphere, and mixed states inside it, in the *Bloch ball*. At the centre lies the maximally mixed state $I/2$, a uniformly random bit. The purity is $\mathrm{Tr}[\rho^2] = (1 + |\vec{r}|^2)/2$: 1 on the sphere, and 1/2 at the centre.

The same decomposition works for any number of qubits, with Pauli strings in place of Pauli matrices:

$$\rho = \frac{1}{2^n} \sum_P \langle P \rangle\, P, \tag{2.2}$$

where the sum runs over all 4^n Pauli strings on n qubits, and $\langle I \rangle = 1$. To see why, note two facts. The product of two Pauli strings is, up to a factor, another Pauli string, which is the identity only if the two are equal. And every Pauli string other than the identity has trace zero. So $\mathrm{Tr}[PQ] = 2^n$ if $P = Q$ and 0 otherwise: the 4^n Pauli strings are orthogonal, and they form a basis of all $2^n \times 2^n$ matrices. Multiplying the right-hand side of Equation 2.2 by a Pauli string Q and taking the trace keeps only the term $P = Q$, and gives $\mathrm{Tr}[Q\rho] = \langle Q \rangle$, as it should. The expectation values of the Pauli strings fix the state, as the Bloch vector does for one qubit, and most of this chapter follows what noise does to them. To describe some of the qubits on their own, we discard the others with the *partial trace*, defined by $\mathrm{Tr}_B[A \otimes B] = \mathrm{Tr}[B]\, A$. In Equation 2.2, it keeps the terms whose Pauli strings act as the identity on the discarded qubits.

2.1.2 Quantum channels

Noise turns one density matrix into another. The maps that describe this are called quantum channels.

Definition 2.2 (Quantum channel)**.** A *quantum channel* maps a density matrix ρ to

$$\Lambda(\rho) = \sum_k K_k \, \rho \, K_k^\dagger, \qquad \text{with} \quad \sum_k K_k^\dagger K_k = I. \tag{2.3}$$

The matrices K_k are its *Kraus operators*.

Each Kraus operator describes one thing that can happen to the qubit, with probability $\mathrm{Tr}[K_k \rho K_k^\dagger]$. The condition on the Kraus operators keeps the trace, since $\mathrm{Tr}[\Lambda(\rho)] = \mathrm{Tr}[\sum_k K_k^\dagger K_k \, \rho] = \mathrm{Tr}[\rho]$, so these probabilities sum to one. Each term $K_k \rho K_k^\dagger$ is also positive semidefinite, so $\Lambda(\rho)$ is again a density matrix. This still holds if the channel acts on only some of the qubits of a larger system, a property called *complete positivity*. It is stronger than positivity: the transpose, for example, is positive but not completely positive. Conversely, every linear map that is completely positive and keeps the trace (CPTP) has a Kraus form. More details, with proofs, can be found in Nielsen and Chuang (2010, sec. 8.2) and in the Quantum Information Theory notes of Walter and Ozols (2025, Theorem 5.3 and Lemma 5.5, where the Kraus operators are called X_i).

Where do channels come from? A qubit is never perfectly isolated (Section 1.2.1): it interacts with its environment, for example with defects in the chip (Section 1.5). Qubit and environment together evolve unitarily, but we do not keep track of the environment, so we discard it with the partial trace. The result is always a channel on the qubit. Conversely, every channel can arise in this way, and the unitary and the environment that produce it are called a *Stinespring dilation* of the channel. Exercise 2.10 works out an example, and more details can be found in Walter and Ozols (2025, sec. 5.2).

A unitary gate U is a channel with the single Kraus operator U. Applying Λ_1 and then Λ_2 gives a channel with the Kraus operators $K_j^{(2)} K_k^{(1)}$. And if Λ_i happens with probability p_i, the mixture $\sum_i p_i \Lambda_i(\rho)$ is a channel with the Kraus operators $\sqrt{p_i}\, K_k^{(i)}$. Most noise in this chapter is of this last kind: with some probability, something goes wrong.

2.2 Noise on one qubit

2.2.1 Pauli channels on the Bloch sphere

The simplest errors are Pauli matrices: X flips the bit, Z flips the sign of the relative phase between $|0\rangle$ and $|1\rangle$, and Y does both.

Definition 2.3 (Pauli channel)**.** A *Pauli channel* on one qubit applies the error $Q \in \{I, X, Y, Z\}$ with probability q_Q:

$$\Lambda(\rho) = q_I \, \rho + q_X \, X\rho X + q_Y \, Y\rho Y + q_Z \, Z\rho Z, \qquad q_I + q_X + q_Y + q_Z = 1.$$

Its Kraus operators are $K_Q = \sqrt{q_Q}\, Q$, for $Q \in \{I, X, Y, Z\}$.

What does a Pauli channel do to the Bloch vector? Conjugating a Pauli matrix by another gives back the same matrix, up to a sign: $QPQ = P$ if $Q = I$ or $Q = P$, and $QPQ = -P$ otherwise, because different Pauli matrices anticommute. The channel therefore maps each Pauli matrix to a multiple of itself, $\Lambda(X) = f_x X$, and likewise for Y and Z, with

$$\begin{aligned} f_x &= q_I + q_X - q_Y - q_Z, \\ f_y &= q_I - q_X + q_Y - q_Z, \\ f_z &= q_I - q_X - q_Y + q_Z, \end{aligned} \tag{2.4}$$

while $\Lambda(I) = I$. By Equation 2.1, the channel multiplies each component of the Bloch vector by its own factor, $r_j \mapsto f_j r_j$, and so each expectation value: $\langle X\rangle \mapsto f_x\langle X\rangle$, and so on. We call f_x, f_y and f_z the *shrinking factors* of the channel. They are the main tool of this chapter: as long as the noise is a Pauli channel, its effect on a measured expectation value is a multiplication. In the Bloch ball, a Pauli channel squeezes the ball into an ellipsoid with semi-axes $|f_x|$, $|f_y|$ and $|f_z|$, centred at the maximally mixed state.

2.2.2 Depolarising and dephasing noise

Two Pauli channels describe most of the noise on a single qubit in this chapter.

Definition 2.4 (Depolarising and dephasing channels)**.** The *depolarising channel* on one qubit is the Pauli channel with $q_X = q_Y = q_Z = p/4$:

$$\Lambda(\rho) = \Big(1 - \frac{3p}{4}\Big)\rho + \frac{p}{4}(X\rho X + Y\rho Y + Z\rho Z).$$

The *dephasing channel* is the Pauli channel with $q_Z = p$:

$$\Lambda(\rho) = (1-p)\,\rho + p\,Z\rho Z.$$

The depolarising channel treats the three axes alike. By Equation 2.4, its factors are $f_x = f_y = f_z = 1 - p$. It shrinks the Bloch ball uniformly towards its centre (Figure 2.1 a), and every expectation value with it: $\langle P\rangle \mapsto (1-p)\langle P\rangle$. By Equation 2.1, the state with the Bloch vector $(1-p)\vec{r}$ is the mixture $(1-p)\rho + p\,I/2$, so

$$\Lambda(\rho) = (1-p)\,\rho + p\,\frac{I}{2}\,: \tag{2.5}$$

with probability p, the depolarising channel replaces the state by the maximally mixed state. This form gives other Kraus operators: $\sqrt{1-p}\,I$ and the four matrices $\sqrt{p/2}\;|i\rangle\langle j|$ with $i, j \in \{0, 1\}$. The same channel can therefore have different Kraus operators. The depolarising channel also commutes with every single-qubit unitary U: $U\Lambda(\rho)U^\dagger = (1-p)\,U\rho U^\dagger + p\,I/2 = \Lambda(U\rho U^\dagger)$, since $U(I/2)U^\dagger = I/2$.

The dephasing channel has the factors $f_x = f_y = 1 - 2p$ and $f_z = 1$. It leaves $\langle Z \rangle$, and with it the probabilities of measuring 0 and 1, unchanged. It shrinks the components of the Bloch vector in the xy plane, which carry the relative phase between $|0\rangle$ and $|1\rangle$. The ball is squeezed towards the z axis (Figure 2.1 b), and for $p = 1/2$ it collapses onto it. It is also called the *phase-flip channel*, after its Kraus form, a Z error with probability p. The name suggests a phase that now and then flips suddenly, but dephasing usually comes from a gradual loss of information about the phase, as we see below.

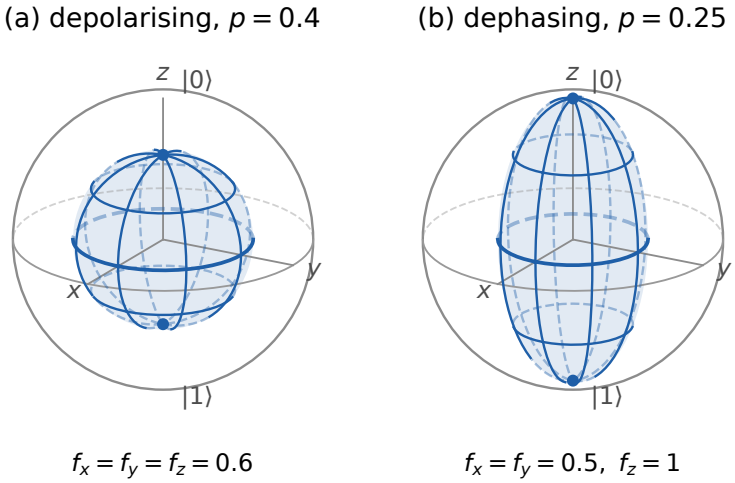


Figure 2.1: The Bloch sphere (grey) and its image (blue) under two Pauli channels. The dots mark where $|0\rangle$ and $|1\rangle$ go. (a) Depolarising noise shrinks the ball uniformly, and (b) dephasing squeezes it towards the z axis.

The physical origin of dephasing. The relative phase of a superposition $\alpha\,|0\rangle + \beta\,|1\rangle$ turns at a rate set by the qubit's frequency (Section 1.5), and the device is calibrated to expect this. Suppose that the frequency of the qubit is shifted by a small amount δ (in radians per second), for example by a fluctuating magnetic field. The qubit then picks up an extra phase that nobody expected: in a time t, its relative phase turns by an extra angle $\theta = \delta t$. This is the rotation $R_Z(\theta)$ of Equation 1.2. If we knew δ, we could undo this rotation. If we do not know it, we do not know θ either, and the qubit undergoes a random rotation. Its state becomes the mixture $\Lambda(\rho) = \mathbb{E}[R_Z(\theta)\,\rho\,R_Z(\theta)^\dagger]$, where $\mathbb{E}$ is the average over θ. By Exercise 1.4, $R_Z(\theta) X R_Z(\theta)^\dagger = \cos\theta\, X + \sin\theta\, Y$. For an angle that is as likely to be positive as negative, the sines average to zero:

$$\Lambda(X) = \mathbb{E}[\cos\theta]\,X, \qquad \Lambda(Y) = \mathbb{E}[\cos\theta]\,Y, \qquad \Lambda(Z) = Z.$$

These are the factors of a dephasing channel with $1 - 2p = \mathbb{E}[\cos\theta]$: an unknown phase acts as dephasing. The longer the qubit waits, the wider the spread of $\theta = \delta t$, and the more information about the phase is lost. Once the spread is much larger than a radian, $\mathbb{E}[\cos\theta]$ is close to zero and p close to $1/2$: the phase is lost. For a Gaussian angle with variance σ^2, $\mathbb{E}[\cos\theta] = e^{-\sigma^2/2}$ (Exercise 2.1).

Exercise 2.1 (Random phases)**.** A qubit undergoes the rotation $R_Z(\theta)$, where θ is a Gaussian random angle with mean zero and variance σ^2.

(a) Show that $\mathbb{E}[\cos\theta] = e^{-\sigma^2/2}$.

Hint: $\cos\theta$ is the real part of $e^{i\theta}$.

(b) Which dephasing probability p does this channel have? What is it for $\sigma \ll 1$?

2.2.3 Noise gate by gate

A common approximation of the noise on a device models each noisy gate as the ideal gate U followed by a channel Λ, so that it maps ρ to $\Lambda(U\rho U^\dagger)$. A circuit of noisy gates is then a composition of channels. Quantum software development kits (SDKs), such as the Qiskit of the tutorials, usually describe noise in this way. The approximation assumes that the noise has no memory: the channel after each gate does not depend on what happened before. Noise of this kind is called *Markovian*. Section 2.2.6 describes the main exception. The shrinking factors make compositions easy: two Pauli channels in a row multiply each expectation value by the factors of both.

Exercise 2.2 (Composing depolarising channels)**.**

(a) Apply the depolarising channel with parameter p_1 and then the one with p_2, using Equation 2.5. Show that the result is again a depolarising channel, and check that its factor is $(1-p_1)(1-p_2)$.

(b) A qubit is prepared in $|+\rangle$, and n noisy X gates are applied, each followed by depolarising noise with parameter p. What is $\langle X\rangle$ at the end? For $p = 10^{-3}$, after how many gates has $\langle X\rangle$ fallen to half of its initial value?

Hint: Where in the circuit can the channels be moved?

(c) Two qubits each undergo depolarising noise with parameter p, independently, so that the channel on both is $\Lambda \otimes \Lambda$. By which factors do $\langle Z \otimes I\rangle$ and $\langle Z \otimes Z\rangle$ shrink? Do these factors depend on the state of the two qubits?

Hint: A product of channels acts factor by factor: $(\Lambda \otimes \Lambda)(A \otimes B) = \Lambda(A) \otimes \Lambda(B)$.

2.2.4 Relaxation and non-unital noise

The channels so far are Pauli channels, and they share a property: they map the maximally mixed state to itself, $\Lambda(I/2) = I/2$, so the centre of the Bloch ball stays where it is. A channel

with this property is called *unital*. Unital noise can only make a state more mixed: it destroys information, but it does not push the qubit towards any particular state.

Relaxation, the decay of $|1\rangle$ to $|0\rangle$ (Section 1.5), is different. A qubit in $|1\rangle$ gives its energy to the environment and ends up in $|0\rangle$. Relaxation therefore pushes every state towards $|0\rangle$, the maximally mixed state included: it is not unital. It is described by the *amplitude-damping* channel, with the Kraus operators

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \qquad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix} = \sqrt{\gamma}\,|0\rangle\langle 1|\,. \tag{2.6}$$

K_1 is the decay itself: a qubit in $|1\rangle$ jumps to $|0\rangle$, with probability γ. K_0 is the case without a decay, which also shrinks the amplitude of $|1\rangle$, since a qubit that has not decayed is more likely to have been in $|0\rangle$ all along. Applied to a density matrix, the channel gives

$$\begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \mapsto \begin{pmatrix} \rho_{00}+\gamma\rho_{11} & \sqrt{1-\gamma}\,\rho_{01} \\ \sqrt{1-\gamma}\,\rho_{10} & (1-\gamma)\,\rho_{11} \end{pmatrix}.$$

In terms of the Bloch vector, $r_z \mapsto (1-\gamma)\,r_z + \gamma$ and $r_{x,y} \mapsto \sqrt{1-\gamma}\,r_{x,y}$: the ball shrinks, and its centre moves towards $|0\rangle$, to $r_z = \gamma$ (Figure 2.2 a). Non-unital noise can thus make a state purer: for $\gamma = 1$, even the maximally mixed state becomes the pure state $|0\rangle$.

2.2.5 Relaxation (T_1) and dephasing (T_2^*) times

Two experiments measure how fast a qubit relaxes and dephases, and define two of the times that providers quote for their devices (Section 1.5).

Definition 2.5 (Relaxation time T_1)**.** Prepare a qubit in $|1\rangle$ and wait for a time t: the probability of still finding it in $|1\rangle$ decays as e^{-t/T_1}. T_1 is called the *relaxation time*.

Definition 2.6 (Dephasing time T_2^*)**.** Prepare a qubit in $|+\rangle$ and wait for a time t: its *coherence*, the size $|\rho_{01}|$ of the off-diagonal entry of its density matrix, decays approximately as e^{-t/T_2^*}. T_2^* is called the *dephasing time*. It is measured with a *Ramsey experiment* (Figure 2.3 a): a $\pi/2$ pulse turns $|0\rangle$ into $|+\rangle$, and after the wait, a second $\pi/2$ pulse turns the x component of the Bloch vector into the z component, which the measurement reads out.

Which channels produce these two laws? For T_1, it is amplitude damping: after a time t, the probability ρ_{11} of finding $|1\rangle$ has decayed by $1-\gamma$, so $\gamma = 1 - e^{-t/T_1}$. Amplitude damping also shrinks the coherence, by $\sqrt{1-\gamma} = e^{-t/2T_1}$. Even without dephasing, relaxation therefore limits the coherence: for a pure state $\alpha\,|0\rangle + \beta\,|1\rangle$, the coherence is $|\alpha\beta|$, and the amplitude β decays at half the rate of the probability $|\beta|^2$.

The rest of the decay is dephasing, caused by the unknown phases of Section 2.2.2. The qubit frequency is shifted by an amount that we do not know, so the phase drifts by an angle that

we do not know, and after some time the information about the phase is lost. We describe this by a dephasing channel whose factor decays as $1 - 2p = e^{-t/T_\varphi}$. This is called *pure dephasing*, to distinguish it from the loss of coherence caused by relaxation. Its time T_φ is called the pure-dephasing time. Amplitude damping and dephasing commute (Exercise 2.3), and the coherence decays with both factors, $e^{-t/2T_1} e^{-t/T_\varphi} = e^{-t/T_2^*}$, where

$$\frac{1}{T_2^*} = \frac{1}{2T_1} + \frac{1}{T_\varphi}. \tag{2.7}$$

Figure 2.2 b shows the combined channel.

Since $1/T_\varphi \geq 0$, Equation 2.7 gives $T_2^* \leq 2T_1$. This bound matters in practice. The dephasing channel needs T_φ, but T_φ cannot be measured on its own, because relaxation cannot be switched off: every measurement of the coherence includes the decay $e^{-t/2T_1}$. Experiments measure T_1 and T_2^*, and T_φ follows from Equation 2.7. A noise model built from measured values therefore needs $T_2^* \leq 2T_1$. Otherwise, $1/T_\varphi$ would be negative, and the dephasing channel would need $p < 0$, which is not a channel. The simulators of SDKs reject or correct such values.

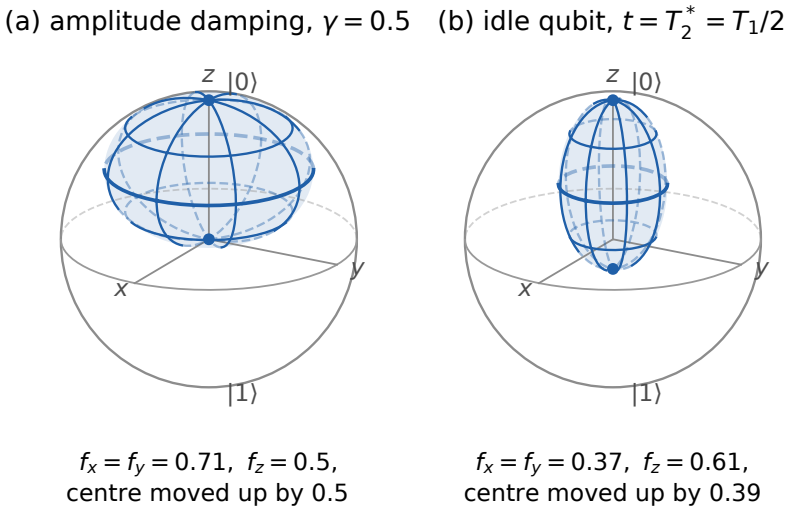


Figure 2.2: The Bloch sphere (grey) and its image (blue) under (a) amplitude damping, which shrinks the ball and pulls it towards $|0\rangle$, and (b) an idle qubit, with relaxation and pure dephasing together, after a wait $t = T_1/2$ with $T_2^* = T_1/2$. The dots mark where $|0\rangle$ and $|1\rangle$ go.

An idle qubit undergoes this combined channel while it waits. For a short wait t, its factors are $1 - t/T_2^*$ on the x and y axes and $1 - t/T_1$ on the z axis, so its error is of order t/T_2^* (Section 2.6.3 makes this precise). A wait of 100 ns with $T_2^* = 100$ µs therefore costs as much as a single-qubit gate, or more. Table 2.1 collects the single-qubit channels.

Table 2.1: Single-qubit channels. The factors multiply the components of the Bloch vector. For an idle qubit, $\gamma = 1 - e^{-t/T_1}$ and $1 - 2p = e^{-t/T_\varphi}$.

Channel	Pauli errors	$f_x = f_y$	f_z	Centre of the ball	Describes
depolarising	X, Y and Z, each with probability $p/4$	$1 - p$	$1 - p$	fixed	gate errors
dephasing	Z with probability p	$1 - 2p$	1	fixed	unknown phases (T_φ)
amplitude damping	not a Pauli channel (Equation 2.6)	$\sqrt{1-\gamma}$	$1 - \gamma$	moves to $r_z = \gamma$	relaxation (T_1)

Exercise 2.3 (Relaxation and dephasing together)**.**

(a) Show that amplitude damping and dephasing commute. Write the Bloch vector of $|+\rangle$ after it has been idle for a time t.

Hint: Compare what the two channels do to ρ_{00}, ρ_{11} and ρ_{01}.

(b) Suppose the calibration data of a superconducting device give, for three of its qubits, $T_1 = 150$, 80 and 200 µs and $T_2^* = 120$, 150 and 60 µs. Compute T_φ for each qubit. Which process limits the coherence of each, relaxation or dephasing?

2.2.6 Echoes and T_2

The dephasing channel describes a phase that we do not know: the frequency shift δ of Section 2.2.2, and with it the angle $\theta = \delta t$, differs from shot to shot by an amount that we cannot predict. Physically, this is a reasonable picture when the frequency fluctuates slowly, so that δ stays constant during each shot. But then the qubit picks up its unknown phase at a steady rate, and there is an easy way to undo it. An X pulse halfway through the wait exchanges $|0\rangle$ and $|1\rangle$, which reverses the sign of the phase picked up so far. The second half of the wait then undoes the first, whatever the value of δ (Exercise 2.4). This is called an *echo*, or Hahn echo, after Erwin Hahn, who found it with nuclear spins in 1950. It leaves the qubit flipped, so a second X is added at the end of the wait. Repeated echoes during a long wait, called *dynamical decoupling*, protect idle qubits from slow noise on the device itself, while the error mitigation of later chapters corrects the estimates after the run.

If the frequency also changes during the wait, the phases picked up in the two halves differ, and the echo cancels only the part they have in common. What is left comes from fluctuations on timescales shorter than the wait. The angle is then a sum of many small, independent kicks, one for each short interval, so it is Gaussian (by the central limit theorem). Its variance is the sum of the variances of the kicks, the fact behind Equation 1.9, and it therefore grows in

proportion to t. We write it as $\sigma^2 = 2t/T_\varphi$, so that the coherence decays as $e^{-\sigma^2/2} = e^{-t/T_\varphi}$. Fast noise gives the exponential decay of pure dephasing that Section 2.2.5 assumed.

An echo experiment, a Ramsey experiment with an X pulse halfway and another at the end of the wait (Figure 2.3), therefore measures a longer time than T_2^*, called T_2: the decay due to relaxation and fast noise only. It obeys Equation 2.7 with T_2 in place of T_2^* and a longer T_φ, so $T_2^* \le T_2 \le 2T_1$. The star of T_2^* marks the time that includes slow noise. Dynamical decoupling is usually assumed to be available for idle qubits, so calibration data often quote T_2 rather than T_2^*, and the error budgets of Section 2.6.3 use T_2 too. In a Ramsey experiment, slow noise makes the coherence decay faster, and the decay need not be exponential: for a frequency shift that is constant within each shot and Gaussian across shots, it is a Gaussian in t (Exercise 2.4).

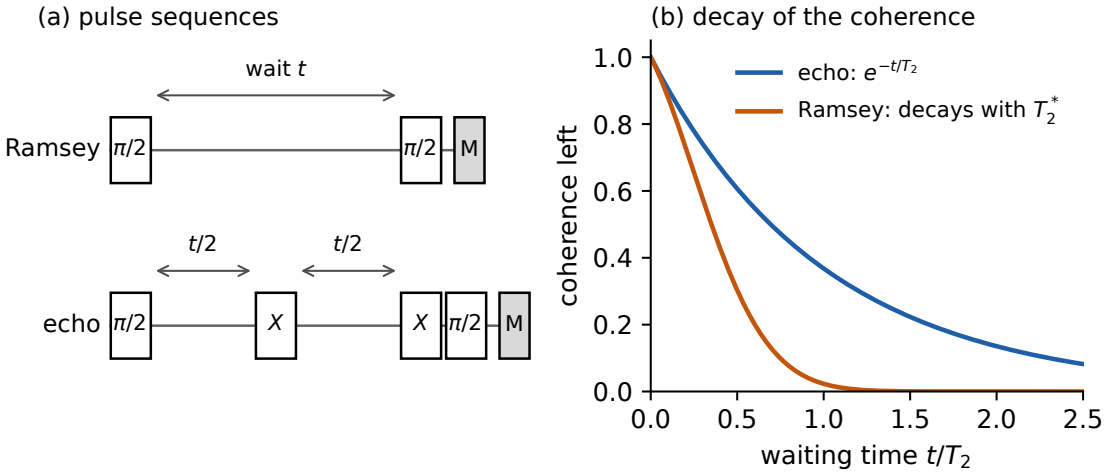


Figure 2.3: (a) A Ramsey experiment and an echo. Each $\pi/2$ pulse is a rotation by $\pi/2$ about an axis in the xy plane, and M is the measurement. (b) The coherence left after a wait t. With slow noise, the Ramsey signal decays faster than the echo signal, which decays as e^{-t/T_2}. The curves are an illustration, with slow noise of one fixed strength.

Slow noise has memory, so it is the main exception to the Markovian model of Section 2.2.3. A common source is a defect in the chip that switches between two states and makes the qubit frequency jump (*random telegraph noise*). Many such defects give noise in which slow fluctuations dominate (*1/f noise*, with f the frequency of the fluctuations).

Exercise 2.4 (The echo)**.**

(a) Show that $XR_Z(\theta)X = R_Z(-\theta)$, and that $R_Z(\theta/2)\,X\,R_Z(\theta/2) = X$ for every angle θ. An echo on an idle qubit ends with a second X. Why does this second X cost nothing extra when the next gate on the qubit is a single-qubit gate?

Hint: Use $XZX = -Z$ and Equation 1.5. For the second X, see Section 1.3.5.

(b) In a Ramsey experiment, the qubit frequency is shifted by an amount δ that is constant within each shot, and Gaussian with standard deviation σ_δ across shots. Ignoring relaxation and fast noise, show that the coherence after a wait t is $e^{-\sigma_\delta^2 t^2/2}$. What is it with an echo?

Hint: Use Exercise 2.1 with $\theta = \delta t$.

2.3 Noise on many qubits

On many qubits, the tools of Section 2.2.1 carry over with one more fact of Pauli algebra. Two Pauli strings P and Q either commute, $PQ = QP$, or anticommute, $PQ = -QP$; we can summarise this by defining the sign $s_{PQ} = \pm 1$ such that $PQ = s_{PQ}\, QP$. A Pauli channel on n qubits, $\Lambda(\rho) = \sum_Q q_Q\, Q\rho Q$, therefore multiplies the expectation value of every Pauli string P by its own factor,

$$\langle P\rangle \mapsto f_P \langle P\rangle, \qquad f_P = \sum_Q s_{PQ}\, q_Q. \tag{2.8}$$

Noise that acts on each qubit separately, with a channel Λ_j on qubit j, is called *local*: the channel on all qubits is $\Lambda_1 \otimes \cdots \otimes \Lambda_n$. It acts factor by factor on a Pauli string and leaves the identity factors unchanged, so the factor of a string is the product of the single-qubit factors at its non-identity positions.

As examples, we extend the simplest noise of Section 2.2.2, the depolarising channel, to many qubits. There are two natural ways to do this.

Definition 2.7 (Local depolarising noise)**.** *Local depolarising noise* with parameter p applies the depolarising channel of Definition 2.4 to each qubit, independently.

Definition 2.8 (Global depolarising noise)**.** The *global depolarising channel* on n qubits replaces the state of all qubits by the maximally mixed state with probability p:

$$\Lambda(\rho) = (1-p)\,\rho + p\,\frac{I}{2^n}.$$

For $n = 1$, both are the depolarising channel of Equation 2.5. With local depolarising noise, a Pauli string of *weight* w, its number of non-identity positions, shrinks by $(1-p)^w$: strings of high weight lose their signal fastest.

Global depolarising behaves differently. Compare the Kraus forms first. With local depolarising noise, each qubit independently suffers X, Y or Z with probability $p/4$ each, so an error on w given qubits has a probability of order $(p/4)^w$: errors on many qubits at once are rare. The

global channel is a Pauli channel too. Averaging $Q\rho Q$ over all 4^n Pauli strings Q gives $I/2^n$. Indeed, by Equation 2.8, the factor of every non-identity string P vanishes: at a position where P is not the identity, two of the four choices of Q anticommute with it, so half of all strings do. So

$$(1-p)\,\rho + p\,\frac{I}{2^n} = \Big(1-p+\frac{p}{4^n}\Big)\rho + \frac{p}{4^n}\sum_{Q\neq I} Q\rho Q.$$

All 4^n-1 non-identity errors are equally likely, and an error on all n qubits is as likely as an error on one. As a result, global depolarising shrinks every Pauli string by the same factor, $1-p$, whatever its weight. No product of local channels does this: it would need $f_{ZI} = f_{IZ} = f_{ZZ} = 1-p$, while a product gives $f_{ZZ} = f_{ZI}\,f_{IZ}$.

Like its single-qubit version, global depolarising noise commutes with every unitary, here on all n qubits. If every gate i of a circuit is followed by global depolarising noise with parameter p_i, all of it can be moved to the end of the circuit. Every expectation value then shrinks by the same factor:

$$\langle P\rangle \mapsto f\langle P\rangle, \qquad f = \prod_i (1-p_i). \tag{2.9}$$

A single number describes the noise of the whole circuit. This makes global depolarising noise much easier to model than local noise, and to correct for (Section 2.7.2).

Local noise does not commute with entangling gates. Conjugation by CZ turns an X on the second qubit into $Z\otimes X$ (Section 1.3.4), so an X error before a CZ is a two-qubit error after it. Errors therefore spread through a circuit: each later two-qubit gate can carry an error to one more qubit, and the qubits that it can reach form a cone, called its *light cone* (Figure 2.4). For deep circuits of random gates, global depolarising noise is then a good approximation of local noise. These circuits are called *scrambling*: the light cone of every error soon covers all the qubits, so that any error, wherever it happens, leaves the output close to random. Global depolarising noise is therefore a very useful model for estimates (Section 2.6 and Section 2.7), although real noise is mostly local.

Correlated errors affect several qubits together. An example is *crosstalk* (Section 1.2.3), including the coupling between neighbouring superconducting qubits that is never fully off and acts as a small, unwanted R_{ZZ} rotation (Equation 1.3). Some correlated errors affect a whole chip. When a cosmic ray, or the radioactive decay of a trace impurity, deposits energy in a superconducting chip, the energy spreads as vibrations of the crystal (*phonons*), and for tens of milliseconds all qubits on the chip relax much faster than usual. In the experiment of McEwen et al. (2022), this happened about once every ten seconds. Such bursts break the independence of errors that the estimates of Section 2.6 assume, and they matter most for error correction.

2.4 Noisy gates

In Section 2.2.3, we modelled the noisy gates of a device as ideal gates, each followed by a channel. But which channel is the right one? Part of it is the relaxation and dephasing during

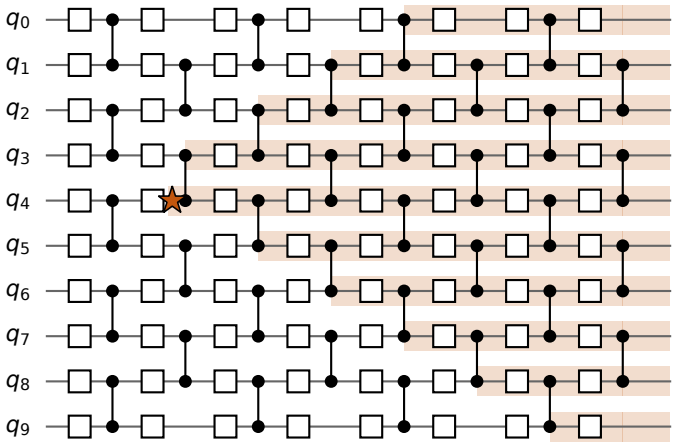


Figure 2.4: One error (star) in the brick-wall circuit of Exercise 1.8, on qubit q_4 after the pulse of the second layer. The shaded wires are its light cone: the qubits that the later CZ gates can carry it to. By the last layers, it can reach every qubit.

the gate. The rest comes from the control: the pulses that implement a gate are never exactly right.

A *coherent error* (Section 1.5) is a fixed, wrong unitary. A pulse that is slightly too strong, for example, implements $R_X(\theta+\eta) = R_X(\eta)R_X(\theta)$ instead of $R_X(\theta)$: the channel after the gate is the unitary $R_X(\eta)$. Biased angles in parametrised gates return with variational algorithms. An *incoherent error* is random: the error unitary differs from shot to shot, and averages to a channel, as the random R_Z did in Section 2.2.2.

The difference matters when a gate is repeated. Apply an X gate n times to a qubit that starts in $|0\rangle$, each time over-rotated about the x axis by a small angle η_k. The result is X^n followed by $R_X(\sum_k \eta_k)$. A total error angle θ gives the wrong outcome with probability $\sin^2(\theta/2)$. We use this *error probability* as the measure of error until Section 2.6. If the error is the same every time, $\eta_k = \eta$, the angles add up like the slow noise of Section 2.2.6, and the error probability is $\sin^2(n\eta/2) \approx n^2\eta^2/4$. If the errors are random and independent, with mean zero and variance σ^2, they add up like the fast noise there: the variance of the total angle is $n\sigma^2$, and the average error probability is about $n\sigma^2/4$. For the same error per gate, $\eta = \sigma$, the coherent error after n gates is n times larger (Figure 2.5).

Three consequences matter later:

1. The error budgets of Section 2.6 assume incoherent errors, so they underestimate coherent ones.
2. Circuits that repeat identical layers, such as those for time evolution, are the worst case.
3. Coherent errors can be made incoherent on purpose: random Pauli gates before and after a gate such as CZ, chosen so that they cancel in the ideal circuit, average its error into a

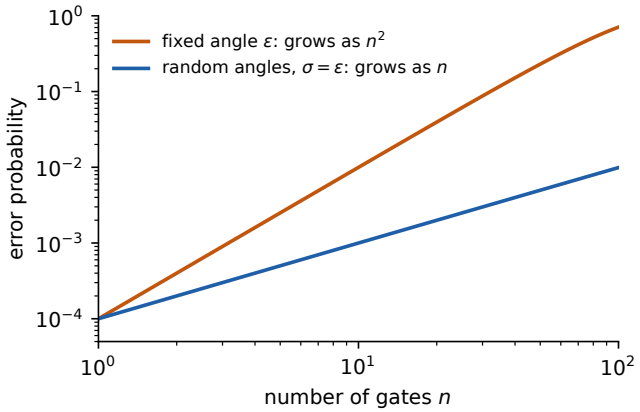


Figure 2.5: The error probability after n over-rotated gates, with the same error per gate: a fixed angle $\eta = 0.02$ (orange) and independent random angles with standard deviation $\sigma = 0.02$ (blue). On these logarithmic axes, the coherent error grows with slope 2 and the incoherent one with slope 1.

Pauli channel. This is called *twirling*. Each random Pauli gate can be merged with the single-qubit gate next to it, which costs a single pulse anyway (Section 1.3.5), so twirling needs no extra pulses. Exercise 2.5 (b) shows the simplest case.

Leakage falls outside this description: it takes a qubit out of its two levels, for example into the third level of a transmon (Figure 1.2), so that channels on 2×2 density matrices cannot describe it.

Exercise 2.5 (Coherent and incoherent errors)**.**

(a) A gate has a coherent over-rotation of $\eta = 0.01$. After how many repetitions is the error probability 1%? Answer the same question for random over- and under-rotations, with mean zero and standard deviation $\sigma = 0.01$.

(b) An $R_X(\theta)$ gate has a small coherent error: it applies $R_Z(\eta)$ after the intended rotation. For each repetition of the gate, independently and with probability 1/2, we add a perfect X gate just before it and another just after it. Since X commutes with $R_X(\theta)$, the ideal gate is unchanged, but the error becomes $XR_Z(\eta)X$. Show that, averaged over these random choices, the error of each gate becomes a dephasing channel with $p = \sin^2(\eta/2)$. How does it now add up over n repetitions?

Hint: Use Exercise 2.4 (a), and compute the factors.

2.5 Readout errors

A measurement can report the wrong bit (Table 1.4). We describe this with the *confusion matrix* M, whose entry $M_{ij} = p(\text{read } i \mid \text{state } j)$ is the probability of reading i when the qubit is in $|j\rangle$. On one qubit,

$$M = \begin{pmatrix} 1 - e_0 & e_1 \\ e_0 & 1 - e_1 \end{pmatrix},$$

where $e_0 = p(1 \mid 0)$ and $e_1 = p(0 \mid 1)$ are the two error rates. If we call reading 1 a positive, e_0 is the rate of false positives and e_1 the rate of false negatives. M maps the probabilities of the true outcomes to those of the recorded ones, so it is a channel on classical probabilities (Walter and Ozols 2025, sec. 4.3).

If the two error rates are equal, $e_0 = e_1 = e$, the readout acts as a perfect measurement after a *bit flip*: the Pauli channel that applies X with probability e. By Equation 2.4, $\langle Z \rangle$ then shrinks by $1 - 2e$, as in Exercise 1.9. Usually, $e_1 > e_0$. A readout on a superconducting qubit takes about a microsecond (Table 1.5), and a qubit in $|1\rangle$ can relax during it, with a probability of order 1% for $T_1 = 100$ µs. With $p_0 = (1 + \langle Z \rangle)/2$ and $p_1 = (1 - \langle Z \rangle)/2$, the recorded expectation value is then

$$\langle Z \rangle_{\text{read}} = (1 - e_0 - e_1)\,\langle Z \rangle + (e_1 - e_0), \tag{2.10}$$

a shrinking factor and an offset, as for amplitude damping.

Preparation can fail too. A qubit that is thermally excited (Exercise 1.1) starts in $|1\rangle$ instead of $|0\rangle$, and a measurement right after the preparation cannot tell this apart from a readout error (Exercise 2.6). Errors in state preparation and measurement are therefore characterised together, as *SPAM errors*.

On n qubits, the confusion matrix is $2^n \times 2^n$: its entry M_{xy} is the probability of reading the bit string x when the qubits are in the basis state $|y\rangle$. For independent readouts, it is the tensor product of single-qubit matrices. If, in addition, each qubit's two error rates are equal, every qubit of a Pauli string shrinks its expectation value by its own factor $1 - 2e_j$ (Exercise 1.9). Otherwise, the offsets of Equation 2.10 also mix in the expectation values of shorter strings. Correcting estimates for known readout errors is the simplest form of error mitigation, which we meet later in the course.

Exercise 2.6 (Unequal readout errors)**.** A qubit has the readout error rates $e_0 = 1\%$ and $e_1 = 4\%$.

(a) Which value of $\langle Z \rangle$ is recorded on average for the states $|0\rangle$, $|1\rangle$ and $|+\rangle$?

(b) Suppose instead that the readout is perfect, but that preparing $|0\rangle$ gives $|1\rangle$ with probability e_p, and that $|1\rangle$ is prepared by a perfect X gate after that. Which confusion matrix does an experiment that prepares $|0\rangle$ and $|1\rangle$ and measures them report? Can it tell this situation apart from a readout error?

(c) Why does the difference between the two situations matter for a circuit with gates between the preparation and the measurement?

Hint: Where in the circuit does each error act, and what do the gates do to it?

2.6 Fidelity and error budgets

Providers summarise the quality of each operation by a fidelity (Section 1.5). In this section, we define fidelities, and we use the error rates that providers quote for a quick estimate of how much signal a circuit keeps. We work out one example, the same circuit on two platforms.

2.6.1 Fidelity of states and gates

Definition 2.9 (Fidelity)**.** The *state fidelity* of a state ρ with a pure target state $|\psi\rangle$ is $F = \langle\psi|\,\rho\,|\psi\rangle$. The *average gate fidelity* F_{avg} of a noisy gate is the state fidelity of its output with the ideal output $U|\psi\rangle$, averaged over all pure input states $|\psi\rangle$. The *error per gate* is $r = 1 - F_{\text{avg}}$.

The state fidelity is the probability that ρ passes a measurement that asks "is the state $|\psi\rangle$?". Some texts, including Nielsen and Chuang (2010) and the Quantum Information Theory notes (Walter and Ozols 2025, sec. 4.2), call $\sqrt{\langle\psi|\,\rho\,|\psi\rangle}$ the fidelity instead. We use the squared version, as Qiskit does.

For example, after depolarising noise with parameter p, every pure state has the fidelity $1 - p/2$ with the ideal one, so $r = p/2$.

Providers usually quote the error per gate r, measured with *randomised benchmarking*. An error budget needs instead the probability that an operation goes wrong. For a Pauli channel, this is the *Pauli error* $\varepsilon = 1 - q_I$, the probability of an error other than the identity. It is slightly larger than r. On d levels ($d = 2^n$ for n qubits),

$$\varepsilon = \frac{d+1}{d}\, r, \tag{2.11}$$

so $\varepsilon = 3r/2$ for a single-qubit gate and $5r/4$ for a two-qubit gate. Research papers often quote Pauli errors instead of r, so a budget must check which of the two it uses. A derivation of Equation 2.11 can be found in Nielsen (2002).

2.6.2 Circuit fidelity and the signal left

A circuit has many gates, each with a small error. Before we run it, we would like to know how much of the ideal result survives: by how much a measured expectation value shrinks, and whether what is left can be seen above the shot noise. Computing this exactly means simulating the noisy circuit, with a detailed noise model. Instead, we estimate it from the error rates that providers quote, with a few approximations.

Suppose every gate and idle step i of a circuit suffers a Pauli error with probability ε_i, independently of the others. The probability that a shot runs without any error is then

$$f = \prod_i (1 - \varepsilon_i) \approx e^{-\sum_i \varepsilon_i}, \tag{2.12}$$

for small ε_i, as in Section 1.1 and Exercise 1.8. Virtual Z gates add nothing to the sum (Section 1.3.5). In a scrambling circuit (Section 2.3), any error leaves the output close to random, so the noise acts as global depolarising noise. Every expectation value then shrinks by the same factor f, as in Equation 2.9 with $p_i \approx \varepsilon_i$. So f, the probability of a shot without an error, is also the fraction of the signal that is left, and approximately the fidelity of the output state with the ideal one.

Readout errors act after the last gate and do not spread. With equal error rates e, they shrink a Pauli string of weight w by $(1-2e)^w \approx e^{-2ew}$ (Exercise 1.9). Each qubit of the string therefore adds $2e$ to the sum in Equation 2.12, and the other qubits add nothing.

These estimates assume independent, incoherent errors and a circuit that spreads them. They are rough, but they are what we use to judge whether a result is plausible. With N shots, the standard error is about $1/\sqrt{N}$ (Equation 1.9). A measured value $f\langle P\rangle$ can therefore be told apart from zero only if $f|\langle P\rangle|$ is several times larger than $1/\sqrt{N}$.

2.6.3 Example: error budgets on two platforms

As an example, we estimate how much signal one circuit keeps on a superconducting and on a trapped-ion device. Take the brick-wall circuit of Exercise 1.8, on a line of 10 qubits with 8 layers, and estimate $\langle Z \otimes Z\rangle$ on two neighbouring qubits. Each layer applies an R_X pulse to every qubit, its R_Z being virtual, and CZs between neighbours: 80 pulses and 36 CZs. We take values that are typical of current devices (Section 1.5 and Table 1.5):

- superconducting: Pauli errors of 0.03% per pulse (40 ns), 0.5% per CZ (80 ns) and 1% per readout, $T_1 = 150$ µs and $T_2 = 100$ µs;
- trapped ions: 0.01% per pulse (20 µs), 0.2% per two-qubit gate (300 µs) and 0.3% per readout, $T_2 = 1$ s and a much longer T_1, with all gates run one after another.

Idle qubits contribute too. Over a short wait t, the error probability of an idle qubit is about $t/(2T_2) + t/(4T_1)$, from the factors of Section 2.2.5. It uses T_2 in place of T_2^*, since dynamical decoupling protects the idle qubits (Section 2.2.6). On the superconducting device, only the two qubits at the ends of the line wait, during the CZs of every other layer. On the ion trap, each qubit waits for about 10 ms of the 12 ms that the gates take. Table 2.2 collects the contributions.

Table 2.2: Error budget of the 8-layer brick-wall circuit on 10 qubits, for $\langle Z \otimes Z \rangle$ on two neighbouring qubits, with the values in the text.

Contribution to $\sum_i \varepsilon_i$	Superconducting	Trapped ions
two-qubit gates	$36 \times 0.5\% = 0.18$	$36 \times 0.2\% = 0.072$
single-qubit pulses	$80 \times 0.03\% = 0.024$	$80 \times 0.01\% = 0.008$
idling	8 waits of 80 ns: 0.004	about 100 ms in total: 0.05
readout of the two qubits	$2 \times 2 \times 1\% = 0.04$	$2 \times 2 \times 0.3\% = 0.012$
total	0.25	0.14
signal left, f	0.78	0.87
duration of the gates	about 1 µs	about 12 ms

On the superconducting device, the two-qubit gates dominate, followed by the readout. Idling hardly matters, since all qubits are busy in most layers. On the ion trap, every error rate is lower, but the gates take long and run one after another. Each qubit therefore waits for most of the shot, and idling costs almost as much as the two-qubit gates. In general, readout errors dominate wide, shallow circuits and observables of high weight, while gate errors dominate deep circuits.

The same specifications define a *noise model* for a simulator: after each gate, a depolarising channel with the relaxation and dephasing of its duration, together matching the quoted error; the same relaxation and dephasing on idle qubits; and a confusion matrix per qubit. Such a model is only as good as the last calibration (Section 1.5). Tutorial 2 builds such a model in code: open the T2 notebook in Colab.

Exercise 2.7 (The largest visible circuit)**.** Take the brick-wall circuit of Table 2.2 with L layers instead of 8, and suppose that the ideal value of $\langle Z \otimes Z \rangle$ is 0.5. We take 10^4 shots and call the signal visible if the measured value is at least three standard errors away from zero.

(a) Write the signal left, f, as a function of L for each platform.

Hint: Split the contributions of Table 2.2 into a part per layer and a fixed part.

(b) What is the largest L with a visible signal on each platform? How long do the gates of the 10^4 shots take in total on the ion trap?

(c) Which contribution limits L on each platform? Which improvement would help the ion trap most?

2.7 Bias and shot noise

2.7.1 Mean squared error

In Section 1.4.3, the sample mean $\bar{s}$ of N shots was an unbiased estimator of $\langle P \rangle$. With noise, each shot comes from the noisy state, and $\bar{s}$ averages to $f\langle P \rangle$ instead. Its error now has two parts. The *bias* is the difference between its average and the value we want, $b = \mathbb{E}[\bar{s}] - \langle P \rangle = (f-1)\langle P \rangle$. Its variance is $(1 - f^2\langle P \rangle^2)/N$, the square of Equation 1.9 with $f\langle P \rangle$ in place of $\langle P \rangle$. The two add up in the *mean squared error*:

$$\mathbb{E}\left[(\bar{s} - \langle P \rangle)^2\right] = b^2 + \mathrm{Var}(\bar{s}) = (1-f)^2\langle P \rangle^2 + \frac{1 - f^2\langle P \rangle^2}{N}. \tag{2.13}$$

To see the first equality, write $\bar{s} - \langle P \rangle = (\bar{s} - \mathbb{E}[\bar{s}]) + b$ and expand the square: the cross term averages to zero. More shots shrink only the variance, while the bias stays.

The two terms are equal at

$$N^* = \frac{1 - f^2\langle P \rangle^2}{(1-f)^2\langle P \rangle^2}.$$

Beyond N^* shots, the bias dominates the error, and more shots hardly help: they estimate the wrong value more precisely. For the superconducting example of Table 2.2, with $f = 0.78$ and $\langle P \rangle = 0.5$, $N^* \approx 70$ shots. A larger circuit has a smaller f, and an even smaller N^*.

Exercise 2.8 (When more shots stop helping)**.**

(a) Show that $N^* \approx (1 - \langle P \rangle^2)/(\langle P \rangle^2 (\sum_i \varepsilon_i)^2)$ when the total error $\sum_i \varepsilon_i$ of the circuit is small.

(b) For the superconducting brick-wall circuit of Table 2.2 with L layers and $\langle P \rangle = 0.5$, compute the exact N^* for $L = 1$, 8 and 30. Why are more shots worth taking on a better device?

(c) Exercise 1.9 (c) compared 10^3 and 10^5 shots. Compute N^* there, and explain the result of that comparison.

2.7.2 Correcting a known shrinking factor

If the shrinking factor f is known, the bias can be removed: $\bar{s}/f$ averages to $\langle P \rangle$. The price is in the variance. Dividing by f multiplies the standard error by $1/f$, so the same precision as

without noise takes $1/f^2 = e^{2\sum_i \varepsilon_i}$ times as many shots. The cost grows exponentially with the total error of the circuit. For the superconducting example, this is a factor of 1.6. For a circuit ten times larger, with $f \approx 0.08$, it is a factor of about 140. This is the simplest method of error mitigation, applied to readout errors in Exercise 1.9 (d).

Knowing f is the hard part. For global depolarising noise, every expectation value shrinks by the same factor, and the error rates of the device give it (Equation 2.12). Readout errors, which act only at the end of the circuit, are another simple case. For generic circuits and noise models, however, each expectation value is changed in its own way, and predicting it means simulating the noisy circuit. A cost in shots of the same kind returns with the other mitigation methods later in the course.

Two questions follow from here. The observables of real applications, such as the energy of a physical system, are sums of many Pauli strings, each estimated with its own shots and its own noise; the next chapter shows how to measure them. And the bias need not be accepted: error-mitigation methods estimate and remove it, at a price in shots.

2.8 More exercises

These exercises combine material from several sections of the chapter.

Exercise 2.9 (GHZ states under local and global noise)**.** The GHZ state $(|0\cdots0\rangle + |1\cdots1\rangle)/\sqrt{2}$ on n qubits has $\langle X\otimes\cdots\otimes X\rangle = 1$, with X on all qubits, and $\langle Z\otimes Z\otimes I\otimes\cdots\otimes I\rangle = 1$, with Z on the first two qubits only.

(a) Each qubit undergoes depolarising noise with parameter p. By which factors do the two expectation values shrink?

(b) Which parameter of the global depolarising channel shrinks $\langle X\otimes\cdots\otimes X\rangle$ by the same factor? What does it then do to $\langle Z\otimes Z\otimes I\otimes\cdots\otimes I\rangle$?

(c) How could measurements of these two expectation values show whether the noise on a GHZ state is closer to local or to global depolarising?

Exercise 2.10 (Dephasing from an environment)**.** A qubit in the state $\alpha|0\rangle + \beta|1\rangle$ interacts with an environment qubit in the state $\cos(\phi/2)|0\rangle + \sin(\phi/2)|1\rangle$ through a CZ gate, after which the environment is discarded.

(a) Show that the qubit undergoes a dephasing channel with $p = \sin^2(\phi/2)$.

Hint: With the environment second, $\mathrm{CZ} = I\otimes|0\rangle\langle 0| + Z\otimes|1\rangle\langle 1|$. Take the partial trace over the environment.

(b) The Kraus operators of this channel are $K_e = (I \otimes \langle e|)\,\mathrm{CZ}\,(I \otimes |E\rangle)$ for $e = 0, 1$, where $|E\rangle$ is the initial state of the environment. Compute them, and compare them with those of Definition 2.3.

2.9 Further reading

Nielsen and Chuang (2010, sec. 8.3) treats the channels of this chapter in the same Kraus form, and the Quantum Information Theory notes (Walter and Ozols 2025, sec. 4.4 and 5.1-5.2) give the general theory. Krantz et al. (2019) describes T_1, T_2, Ramsey and echo experiments on superconducting qubits.

3 Measuring and simulating Hamiltonians

In the first two chapters, we estimated the expectation values of single Pauli strings on noisy devices. In this chapter, we turn to the observables and the dynamics of physical systems. Both are described by a Hamiltonian, the matrix of the energy of a system, which on a quantum computer becomes a sum of many Pauli strings. These sums stay manageable because physical Hamiltonians are local: each term acts on a few nearby qubits.

By the end of the chapter, you should be able to write a Hamiltonian as a sum of Pauli strings; estimate its energy with a given number of shots and few measurement settings; build circuits for time evolution from native gates and count their cost; and weigh the error of such a circuit against the noise of its gates.

3.1 Energy and time evolution

The *Hamiltonian* H of a system is the observable representing its energy. On n qubits, it is a Hermitian $2^n \times 2^n$ matrix. The eigenvalues E_k of H are the possible outcomes that a measurement of the energy can give, and its eigenvectors $|E_k\rangle$ are the states with a definite energy. Together, they solve the time-independent *Schrödinger equation*

$$H\,|E_k\rangle = E_k\,|E_k\rangle\,.$$

The lowest eigenvalue is the ground-state energy, and its eigenvector the *ground state*. They matter because a system that is cold enough settles in its ground state (or very close to it). The ground state of a Hamiltonian modeling a material determines how that material behaves at low temperature, for example whether it is magnetic, or whether it is superconducting. Differences between energies decide much of chemistry: whether a molecule is stable, how much energy a chemical reaction needs, which frequency of light a molecule absorbs (and hence its color). Finding ground states is the subject of the next chapter.

The Hamiltonian also decides how a state changes in time, through the time-dependent Schrödinger equation and its solution,

$$i\,\frac{d}{dt}\,|\psi(t)\rangle = H\,|\psi(t)\rangle\,, \qquad |\psi(t)\rangle = e^{-iHt}\,|\psi(0)\rangle\,. \tag{3.1}$$

We use units in which the reduced Planck constant $\hbar$ is 1: energies and frequencies are measured in the same unit, and times are measured in units of one over energy.[1] Since H is Hermitian, e^{-iHt} is unitary. In the eigenbasis of H, $e^{-iHt} = \sum_k e^{-iE_k t} |E_k\rangle \langle E_k|$: each eigenstate only picks up a phase, at a rate set by its energy, and a superposition changes because these phases turn at different rates.

For a single qubit with $H = -\frac{\omega}{2} Z$, for example, the relative phase of $|0\rangle$ and $|1\rangle$ turns at the angular frequency $\omega = E_1 - E_0 = 2\pi f_{01}$, where f_{01} is the qubit frequency of Section 1.2.2. As long as the frequency f_{01} is known, this rotation about the Z axis can be compensated using virtual Z rotations (Section 1.3.5).

3.2 Local Hamiltonians

A Hamiltonian on n qubits is a Hermitian $2^n \times 2^n$ matrix, described by 4^n real numbers. On 50 qubits, that is far too many to write down or store in memory. In practice, physics spares us this cost through *locality*. Interactions in physical systems involve only a few sub-systems (e.g. particles) at a time, often only neighbouring ones, so the Hamiltonian is a sum

$$H = \sum_{i=1}^{M} h_i \tag{3.2}$$

of terms h_i that each act on a few qubits, and as the identity on all others, with a number of terms M that grows polynomially in n rather than as 4^n. Figure 3.1 shows the two common forms of locality.

Definition 3.1 (k-local Hamiltonian)**.** A Hamiltonian $H = \sum_i h_i$ on n qubits is *k-local* if each term h_i acts on at most k qubits, with k fixed as n grows.

Definition 3.2 (Geometrically local Hamiltonian)**.** Place the n qubits on the sites of a chain or a lattice. A Hamiltonian $H = \sum_i h_i$ is *geometrically local* if each term h_i acts only on qubits within a fixed distance of each other, with the distance fixed as n grows.

In the models of this chapter, the terms act on single qubits or on pairs of neighbours. Every qubit then appears in only a few terms, and the number of terms is proportional to n. A geometrically local Hamiltonian is also k-local, but not the other way round: a Hamiltonian with a ZZ term on every pair of qubits is 2-local, but it has $n(n-1)/2$ terms, and no arrangement of the qubits on a lattice makes it geometrically local. Several results below, such as the growth of the Trotter error and the light cones (which we discuss in Section 3.6), rely on this stronger property.

[1] Some texts, including the Quantum Computing notes of de Wolf (2026, sec. 9.1), drop the minus sign for convenience and take Hamiltonian simulation to mean implementing e^{+iHt}, which is evolution under $-H$ and costs the same.

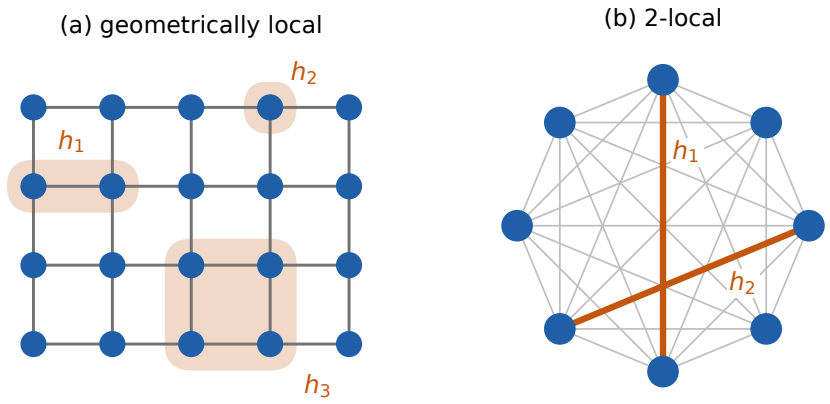


Figure 3.1: Two forms of locality. (a) Geometrically local: the qubits sit on a lattice, and each term acts on qubits close to each other, such as a pair of neighbours (h_1), a single qubit (h_2) or a square of four (h_3). (b) 2-local: each term acts on at most two qubits, but any pair can interact (grey lines), however far apart.

3.2.1 Pauli decomposition

To measure a Hamiltonian, or to build a circuit from it, we write it as a sum of *Pauli strings*. We write $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$ for the set of the 4^n Pauli strings on n qubits (Definition 1.1). They are orthogonal, $\mathrm{Tr}[PQ] = 2^n$ if $P = Q$ and 0 otherwise, and they form a basis of all $2^n \times 2^n$ matrices (Section 2.1.1). Every Hamiltonian can therefore be written as

$$H = \sum_{P \in \mathcal{P}_n} c_P P, \qquad c_P = \frac{\mathrm{Tr}[PH]}{2^n}, \tag{3.3}$$

with real coefficients, since $c_P^* = \mathrm{Tr}[(PH)^\dagger]/2^n = \mathrm{Tr}[HP]/2^n = c_P$.

In general, all 4^n coefficients can be nonzero. For a k-local Hamiltonian, most of them vanish: each term h_i acts as I outside its qubits, so its Pauli strings do too, and all strings have *weight* at most k. Counting the strings of weight $w \leq k$ (choose w qubits, and X, Y or Z on each) shows that a k-local Hamiltonian is a sum of at most $\sum_{w \leq k} \binom{n}{w} 3^w$ Pauli strings, which grows at most like n^k.

3.3 Spin models

Which local Hamiltonians do we study on a quantum computer? The simplest are spin models. Many particles, such as electrons, carry a *spin*, which makes each of them a tiny magnet. The spin of an electron is a *spin-1/2*: measured along any axis, it gives one of two values, up or down: it is a qubit. The Pauli matrices X, Y and Z represent its components along the three

axes (up to a factor $\hbar/2$, which is $1/2$ in our units). In a magnetic material, the spins of neighbouring atoms interact. A spin model describes this with one qubit on each site of a chain or lattice, and terms that couple neighbouring qubits (for spin-1/2; larger spins have more levels). Spin models are standard test problems for quantum computers: each spin is one qubit, all terms are local, and yet many of their properties, in particular their dynamics, are hard to compute classically on large lattices. The Hamiltonians of molecules can be written as sums of Pauli strings too, after an encoding of the electrons into qubits, but with many more terms; we stay with spin models.

Definition 3.3 (Transverse-field Ising model)**.** The *transverse-field Ising model* (TFIM) on a chain or lattice of n qubits is

$$H = -J \sum_{\langle i,j \rangle} Z_i Z_j \; - \; h \sum_{i=1}^{n} X_i,$$

where Z_i is Z on qubit i and the identity on the others, the first sum runs over the *bonds* $\langle i, j \rangle$, the pairs of neighbouring qubits, and $J, h > 0$ are energies. On an open chain, the bonds are the pairs $(i, i+1)$ for $i = 1, \ldots, n-1$.

Each term favours something different. A bond term $-JZ_iZ_j$ has the energy $-J$ when the two qubits agree, in $|00\rangle$ or $|11\rangle$, and $+J$ when they differ: it favours neighbours that are aligned along z. A field term $-hX_i$ has its lowest energy in $|+\rangle$: it favours each qubit pointing along x, transverse to z. No state minimises both kinds of terms at once, since Z and X do not commute. For $h = 0$, the ground states are $|0 \cdots 0\rangle$ and $|1 \cdots 1\rangle$, and for $J = 0$, it is $|+ \cdots +\rangle$. In between, the ground state is entangled, and on a long chain its character changes abruptly at $h = J$, a *quantum phase transition*. Exercise 3.2 works out the smallest case.

On a chain of n qubits, the TFIM is a sum of $2n - 1$ Pauli strings: $n - 1$ bonds and n fields. On an $L \times L$ square lattice (Figure 3.2), it has $2L(L-1)$ bonds and L^2 fields: 280 terms for $L = 10$, against up to 4^{100} for a general Hamiltonian on 100 qubits.

The second model of this chapter describes the magnetism of many materials.

Definition 3.4 (Heisenberg model)**.** The *Heisenberg model* on a chain or lattice of n qubits is

$$H = J \sum_{\langle i,j \rangle} (X_iX_j + Y_iY_j + Z_iZ_j), \tag{3.4}$$

with the bonds $\langle i, j \rangle$ of Definition 3.3 and a coupling energy J.

Each bond treats the three axes alike, so its energy depends only on how the two spins are oriented relative to each other. In the rest of the chapter, the TFIM is the running example, and the Heisenberg model appears where it makes a point more clearly.

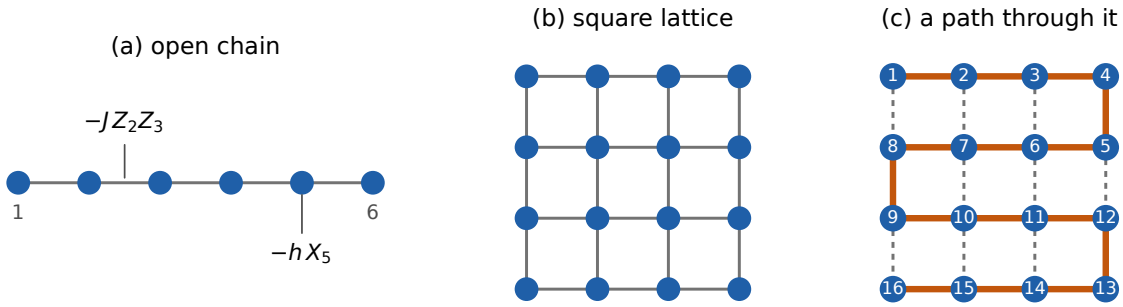


Figure 3.2: Spin models on (a) an open chain of 6 qubits and (b) a 4×4 square lattice. Each circle is a qubit, with its field term $-hX_i$, and each line a bond, with its term $-JZ_iZ_j$ in the TFIM. (c) A path through all qubits of the lattice (orange). Placed on a line of qubits in this order, the bonds along the path join neighbours, while the others (dashed) join qubits up to 7 positions apart (Section 3.5.3).

One property of the Heisenberg model is used for the light cones of Section 3.6. A bond term equals $J(2\,\mathrm{SWAP} - I)$ (Exercise 3.1), so it can move a 1 from one qubit to its neighbour, but never create or remove one. The number of qubits in $|1\rangle$, which we call the number of *flipped qubits*, therefore never changes during the evolution: it is a *conserved quantity* of the model. In particular, $|0\cdots0\rangle$ is an eigenstate.

Exercise 3.1 (Pauli decompositions)**.**

(a) Write $M = \begin{pmatrix} 0 & 1-i \\ 1+i & 2 \end{pmatrix}$ as a sum of Pauli matrices.

(b) Write the SWAP gate as a sum of two-qubit Pauli strings, and conclude that $X \otimes X + Y \otimes Y + Z \otimes Z = 2\,\mathrm{SWAP} - I$. Which eigenvalues does a bond term of the Heisenberg model have, and for which states?

Hint: Use Equation 3.3, and compute each trace as a sum over the basis states $|ab\rangle$.

(c) Show that $e^{-i\theta\,\mathrm{SWAP}} = \cos\theta\, I - i\sin\theta\,\mathrm{SWAP}$. What does this gate do to $|10\rangle$?

Hint: Check which property of P the derivation of Equation 1.1 used.

Exercise 3.2 (The TFIM on two qubits)**.** The *TFIM* on two qubits is $H = -JZ_1Z_2 - h(X_1 + X_2)$.

(a) Write H as a 4×4 matrix in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

(b) Show that H maps the states $|00\rangle + |11\rangle$ and $|01\rangle + |10\rangle$ into combinations of each

other, and the states $|00\rangle - |11\rangle$ and $|01\rangle - |10\rangle$ to multiples of themselves. Find the four eigenvalues, and show that the ground-state energy is $-\sqrt{J^2 + 4h^2}$.

(c) What are the ground states for $h = 0$ and for $J = 0$? Is the ground state entangled for $J = h$?

3.4 Measuring observables

We now estimate the energy $\langle H \rangle$ of a state that a circuit prepares. The examples use the Hamiltonians of Section 3.3, but everything in this section holds for any observable written as a sum of Pauli strings. Local observables, like local Hamiltonians, have few terms. An example is the *magnetisation* $m = \frac{1}{n}\sum_i \langle Z_i \rangle$, the average of $\langle Z_i \rangle$ over the qubits, which measures how far the spins are aligned along z.

We know from Section 1.4.2 how to estimate the expectation value of one Pauli string: rotate each qubit so that its factor becomes Z, measure, and average the signs $s = \pm 1$ of the *shots*. With N shots, the estimate has the standard error $\sqrt{(1 - \langle P \rangle^2)/N}$ (Equation 1.9). For a sum $H = \sum_i c_i P_i$ of M Pauli strings, expectation values are linear, $\langle H \rangle = \sum_i c_i \langle P_i \rangle$, so we can estimate each term and add up the estimates.

3.4.1 The shot budget

Suppose first that each term P_i is measured independently from the others, using N_i shots. The estimates $\bar{s}_i$ of different terms are then independent, so their variances add up, each multiplied by c_i^2:

$$\mathrm{Var}\Big(\sum_i c_i\, \bar{s}_i\Big) = \sum_i \frac{c_i^2 \sigma_i^2}{N_i}, \qquad \sigma_i^2 = 1 - \langle P_i \rangle^2, \tag{3.5}$$

where σ_i^2 is the variance of a single shot of term i. Before the experiment, we do not know the state, so we plan for the worst case, $\sigma_i^2 \le 1$.

How should a total of N shots be split among the terms? We minimise the variance under the constraint $\sum_i N_i = N$, treating the N_i as continuous. With a Lagrange multiplier λ, we can instead minimise $\sum_i c_i^2 \sigma_i^2 / N_i + \lambda \sum_i N_i$ with each N_i free, and choose λ at the end so that the constraint holds. Setting the derivative with respect to N_i to zero gives $c_i^2 \sigma_i^2 / N_i^2 = \lambda$, so N_i is proportional to $|c_i|\sigma_i$. With the constraint,

$$N_i = N \frac{|c_i|\sigma_i}{\sum_j |c_j|\sigma_j}, \qquad \mathrm{Var} = \frac{1}{N}\Big(\sum_i |c_i|\sigma_i\Big)^2.$$

A term gets shots in proportion to $|c_i|\sigma_i$, the spread of its contribution to the energy. Since the σ_i are not known in advance, we can split by $|c_i|$ alone (the worst case), or estimate the σ_i from a first batch of shots. To reach a standard error ε, we need

$$N = \frac{\left(\sum_i |c_i|\sigma_i\right)^2}{\varepsilon^2} \;\le\; \frac{\left(\sum_i |c_i|\right)^2}{\varepsilon^2} \tag{3.6}$$

shots in total. In the worst case, the cost depends only on the sum of the absolute values of the coefficients, the *Pauli 1-norm* $\|c\|_1 = \sum_i |c_i|$ of the vector $c = (c_1, \dots, c_M)$, so that $N \le \|c\|_1^2/\varepsilon^2$. In the literature, $\|c\|_1$ is often written λ; here that letter is already taken by the Lagrange multiplier.

If we instead split the shots equally, $N_i = N/M$, Equation 3.5 gives $\mathrm{Var} = (M/N)\sum_i c_i^2\sigma_i^2 \le M\|c\|_2^2/N$, with the *Pauli 2-norm* $\|c\|_2 = \left(\sum_i c_i^2\right)^{1/2}$. By the Cauchy-Schwarz inequality, $\|c\|_1^2 \le M\|c\|_2^2$, so the optimal split never needs more shots than the equal one. The two are equal when all $|c_i|$ are equal, and the optimal split saves most when a few large coefficients dominate many small ones.

> *Example.* For the *TFIM* on a chain of $n = 10$ qubits with $J = h = 1$, $\|c\|_1 = 19$, and a standard error of 0.1 takes at most $19^2/0.1^2 = 36{,}100$ shots, spread over 19 different circuits. All 19 coefficients have the same size, so in the worst case the equal split is already optimal. Switching between circuits costs time (Section 1.6), so the number of circuits matters too.

3.4.2 Measurement settings

The shots of one circuit can serve several terms at once. In Exercise 1.10, $\langle X \otimes I\rangle$ and $\langle X \otimes X\rangle$ could be estimated from the same shots: after a Hadamard on each qubit, every shot gives both signs. This works whenever two strings need the same rotation on every qubit on which both act.

Definition 3.5 (Qubit-wise commuting strings and measurement settings)**.** Two Pauli strings are *qubit-wise commuting* if, on every qubit, their factors are equal or at least one of them is I. A *measurement setting* assigns a basis, X, Y or Z, to each qubit. It measures every Pauli string whose factors agree with it wherever they are not I, and these strings are qubit-wise commuting.

> *Example.* The *TFIM* needs only two settings, on any chain or lattice: in the Z setting, each shot gives the signs of all bond terms, and in the X setting, with a Hadamard on every qubit, the signs of all field terms (Figure 3.3). The Heisenberg model needs three, X, Y and Z on all qubits, also on any lattice. Its terms X_1X_2 and Y_1Y_2 commute as matrices, but not qubit-wise, so no setting measures both; with an entangling circuit before the measurement, they could be measured together (Section 3.9).

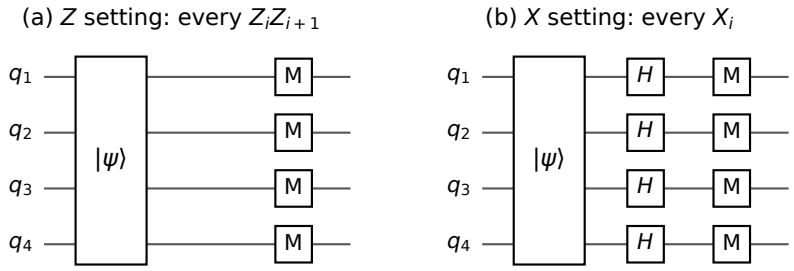


Figure 3.3: The two measurement settings of the TFIM on four qubits, after the circuit that prepares the state $|\psi\rangle$. M is a measurement in the computational basis. (a) Each shot gives the signs of all Z_iZ_{i+1}; (b) after a Hadamard on every qubit, each shot gives the signs of all X_i.

Sharing shots changes the variance. A shot of the setting for a group g of terms gives one value of the whole group, $s_g = \sum_{i\in g} c_i s_i$, and the mean of these values estimates $\sum_{i\in g} c_i\langle P_i\rangle$. The terms now come from the same shots, so the single-shot variance of the group includes their covariances, which measure how two signs vary together (Blitzstein and Hwang 2019). The product s_is_j of two signs from the same shot is the sign of the Pauli string P_iP_j, so that:

$$\sigma_g^2 = \mathrm{Var}(s_g) = \sum_{i,j\in g} c_ic_j\,\mathrm{Cov}(s_i,s_j), \qquad \mathrm{Cov}(s_i,s_j) = \langle P_iP_j\rangle - \langle P_i\rangle\langle P_j\rangle. \tag{3.7}$$

Different settings can use different numbers of shots, so Equation 3.6 holds with groups in place of terms: $N = (\sum_g \sigma_g)^2/\varepsilon^2$ (Exercise 3.4). In the worst case, nothing is gained: $|s_g| \le \sum_{i\in g}|c_i|$, so $\sigma_g \le \sum_{i\in g}|c_i|$, and the bound of Equation 3.6 stays the same. In practice, most pairs of terms in a group act on distant qubits, and in many states distant qubits are nearly uncorrelated. If all terms of a group are uncorrelated, $\sigma_g^2 = \sum_{i\in g} c_i^2\sigma_i^2 \le \sum_{i\in g} c_i^2$: a group of m terms with equal coefficients costs up to m times fewer shots than in the worst case.

> *Example.* For the TFIM chain with $n = 10$, uncorrelated groups give $(\sqrt{9}+\sqrt{10})^2/0.1^2 \approx 3{,}800$ shots instead of 36,100, in 2 circuits instead of 19. For the state that the chain reaches from $|0\cdots0\rangle$ after a time $t = 1$, the exact group variances give about 3,100 shots.

With data, the sample variance of the values s_g of each setting gives the standard error.

Exercise 3.3 (Measurement settings)**.** How many measurement settings does $H = Z_1Z_2 + Z_2Z_3 + X_1 + X_2 + X_3 + Z_1X_2$ need, with single-qubit rotations only? Give the settings and the terms that each measures.
Hint: Look for terms that cannot pairwise share a setting.

Exercise 3.4 (Shots for groups)**.** A Hamiltonian is measured in several settings, with single-shot variances σ_g^2.

(a) Show that the best split of N shots gives setting g a number of shots proportional to σ_g, and that $N = (\sum_g \sigma_g)^2/\varepsilon^2$ shots give the standard error ε.

(b) For the *TFIM* chain with $n = 10$, $J = 1$ and $h = 0.2$, assume that the terms of each group are uncorrelated and $\sigma_i = 1$. How many shots go to each setting for a standard error of 0.1? How many would an equal split need?

(c) A study reports the energy of a state of this chain with a standard error of 0.1 from 200 shots in total. Is the claim plausible? What would have to be true of the state?

3.5 Trotter circuits

We now turn to dynamics: we prepare a simple state, such as $|0\cdots0\rangle$, let it evolve under H for a time t, and measure. In this section, we build circuits for e^{-iHt}.

3.5.1 Simulating dynamics classically

For a few qubits, $e^{-iHt}\,|\psi(0)\rangle$ can be computed exactly on a classical computer. The cost grows exponentially: the state has 2^n complex amplitudes of 16 bytes each, so 30 qubits take about 17 GB of memory, and 50 qubits about 18 PB (1.8×10^{16} bytes), close to the entire memory of the largest supercomputers.

Approximate methods avoid storing all amplitudes by using the structure of the state. For dynamics, the most widely used are *tensor networks*. They write the state as a network of small arrays, one per qubit, connected along the lattice, and the size of the arrays is set by the entanglement between the parts of the system. Product states and states with limited entanglement are therefore cheap to represent. Tensor networks are especially effective at describing dynamics for short times, in particular for *geometrically local* Hamiltonians on a chain. During an evolution, however, the entanglement typically grows in proportion to the time, and the size of the arrays grows exponentially with it, so that long times and two-dimensional lattices stay out of reach. Other methods use other structure, such as the light cones of Section 3.6.

A quantum computer needs no amplitudes in memory: it holds the state in its n qubits and applies e^{-iHt} as a circuit. Simulating quantum systems is the application that motivated the early proposals of quantum computers, such as Feynman (1982). This does not make a quantum computer the right tool for every simulation. For many questions, approximate classical methods do well, on classical hardware that is faster, cheaper and free of noise. A quantum computer is worth using only where all classical methods fail, as for long times or

two-dimensional lattices, and only if its own noise allows (Section 3.7). Finding such problems is part of the work, and it is hard: there is no general test that rules out every classical method, so a claimed *quantum advantage* holds only as long as new classical methods, often developed in response to it, fail to match it.

3.5.2 The Trotter formula

If all terms of H commuted, e^{-iHt} would split exactly into a product of exponentials of single terms, each a rotation about a Pauli string (Equation 1.1). For the *TFIM*, we split $H = A + B$ into the bond terms $A = -J\sum_i Z_i Z_{i+1}$ and the field terms $B = -h\sum_i X_i$. The terms of A commute with each other, and so do those of B, so $e^{-iA\delta}$ and $e^{-iB\delta}$ are products of rotations that a circuit can apply. But A and B do not commute, so applying first $e^{-iA\delta}$ and then $e^{-iB\delta}$, which is the product $e^{-iB\delta}e^{-iA\delta}$ (the rightmost factor acts first), differs from $e^{-i(A+B)\delta}$. For a short time δ, the difference is small. Expanding both sides to second order in δ,

$$\begin{aligned} e^{-iB\delta}e^{-iA\delta} &= I - i(A+B)\delta - \tfrac{1}{2}(A^2 + 2BA + B^2)\delta^2 + O(\delta^3), \\ e^{-i(A+B)\delta} &= I - i(A+B)\delta - \tfrac{1}{2}(A^2 + AB + BA + B^2)\delta^2 + O(\delta^3), \end{aligned}$$

so that

$$e^{-iB\delta}e^{-iA\delta} = e^{-i(A+B)\delta} + \frac{\delta^2}{2}[A,B] + O(\delta^3), \tag{3.8}$$

with the commutator $[A,B] = AB - BA$. The error of one short step is of order δ^2, and it vanishes if A and B commute. To evolve for a time t, we divide it into r steps of length $\delta = t/r$: r errors of order $(t/r)^2$ add up to an error of order t^2/r, which falls as the number of steps grows.

To make this precise, we need a size for matrices. The *operator norm* $\|M\| = \max_{\|\psi\|=1} \|M|\psi\rangle\|$ is the largest factor by which M can stretch the length of a vector; for a Hermitian matrix, it is the largest absolute value of its eigenvalues. For a Pauli sum $H = \sum_i c_i P_i$, the triangle inequality $\|M+N\| \le \|M\| + \|N\|$ and $\|P_i\| = 1$ give $\|H\| \le \|c\|_1$: the *Pauli 1-norm* of Section 3.4.1 also bounds the energies of H.

Definition 3.6 (First-order Trotter formula)**.** For $H = A + B$ and a time t divided into r *Trotter steps* of length $\delta = t/r$,

$$e^{-iHt} \approx \left(e^{-iB\delta}e^{-iA\delta}\right)^r,$$

with an error, the *Trotter error*, of at most

$$\left\| e^{-iHt} - \left(e^{-iB\delta}e^{-iA\delta}\right)^r \right\| \le \frac{t^2}{2r}\,\|[A,B]\|. \tag{3.9}$$

The bound is the same for the other order, $e^{-iA\delta}e^{-iB\delta}$.

Equation 3.9 says that, for every initial state, the state of the circuit differs from the exact one by a vector of length at most $(t^2/2r)\|[A,B]\|$. This bound rests on two facts. First, the error of a single step is at most $(\delta^2/2)\|[A,B]\|$ for every δ. Equation 3.8 suggests this, but the proof takes more work (Childs et al. 2021, sec. V.A). Second, the errors of the r steps add up at most linearly. With $U = e^{-iB\delta}e^{-iA\delta}$ for one Trotter step and $V = e^{-iH\delta}$ for one exact step, the sum

$$V^r - U^r = \sum_{k=0}^{r-1} V^{r-1-k}\,(V-U)\,U^k$$

telescopes: neighbouring terms cancel, and only V^r and $-U^r$ remain. Unitaries do not change the length of vectors, so each term has norm at most $\|V-U\|$, and $\|V^r - U^r\| \leq r\,\|V-U\| \leq r\,(\delta^2/2)\|[A,B]\| = (t^2/2r)\|[A,B]\|$. de Wolf (2026, sec. 9.2) derives the bound $O(\|A\|\|B\|\,t^2/r)$ in the same way.

In general, a Hamiltonian may need more than two groups of commuting terms; each step then applies one exponential per group, and the bound becomes a sum over the commutators of all pairs of groups.

> *Example.* For the TFIM, two groups suffice on any lattice, since all ZZ terms commute with each other, and so do all X terms. The TFIM is also geometrically local (Definition 3.2), so only terms that overlap fail to commute: Z_iZ_{i+1} anticommutes with X_i and X_{i+1}, and commutes with every other X_j. As a result, $\|[A,B]\| \leq 4Jh(n-1)$ (Exercise 3.5): the bound grows in proportion to the number of qubits. The weaker bound $O(\|A\|\|B\|\,t^2/r)$ grows faster: here the norms reach the Pauli 1-norms of the two groups, $\|A\| = J(n-1)$ and $\|B\| = hn$ (all spins aligned along z, or along x), so $\|A\|\|B\|$ grows as n^2.

3.5.3 Circuits for one step

For any Hamiltonian, each group of commuting terms gives one Pauli rotation per term (Equation 1.1), so a Trotter step is a circuit of Pauli rotations. To run it, we arrange the rotations so that those on different qubits act at the same time, compile them into native gates, and fit them on the coupling graph of the device.

> *Example.* For the *TFIM*, these rotations are $e^{iJ\delta Z_iZ_{i+1}} = R_{ZZ}(-2J\delta)$ on every bond, and $e^{ih\delta X_i} = R_X(-2h\delta)$ on every qubit. Two bond gates that share a qubit cannot act at the same time, so the bonds are applied in two layers, first $(1,2)$, $(3,4)$, …, then $(2,3)$, $(4,5)$, …, and the field rotations in a third (Figure 3.4). Two-qubit gates on alternating pairs of neighbours form a *brick-wall circuit*, as in Figure 1.12. On a device with CZ gates, each R_{ZZ} is two CNOTs and an R_Z (Figure 3.4 b), so it costs two CZs, and a step on a chain of n qubits costs $2(n-1)$ CZs; on some trapped-ion devices, $R_{ZZ}(\theta)$ is a single *native gate* (Table 1.3).

A Trotter circuit repeats the same step r times, so a *coherent error* of a gate, such as an over-rotation, repeats r times too and adds up: this is the worst case of Section 2.4.

The lattice of the model must also fit the *coupling graph* of the device (Section 1.3.6). A chain fits on any device whose coupling graph contains a path of n qubits, and every bond is then a

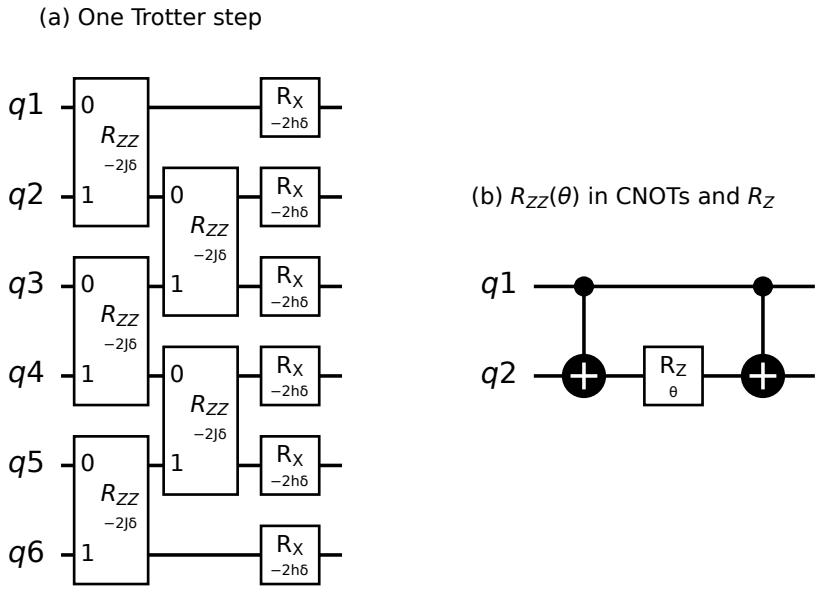


Figure 3.4: One Trotter step of the TFIM on a chain of 6 qubits. (a) $R_{ZZ}(-2J\delta)$ on the bonds $(1,2)$, $(3,4)$, $(5,6)$, then on $(2,3)$, $(4,5)$, then $R_X(-2h\delta)$ on every qubit. (b) Each $R_{ZZ}(\theta)$, here on the bond $(1,2)$, compiled into two CNOTs and one $R_Z(\theta)$, as in Figure 1.6.

coupled pair. A square lattice fits directly on a device with a square coupling graph, such as IQM's 20-qubit Garnet, with four layers of bond gates per step instead of two. On a line of qubits, we can place the lattice along a path through it (Figure 3.2 c). The bonds along the path then join neighbours, but the others join qubits up to $2L-1$ positions apart, and each of them needs *SWAPs* (Figure 1.11), which multiply the cost of a step (Exercise 3.6). Trapped ions, coupled all to all, need no SWAPs.

Exercise 3.5 (The commutator of the TFIM)**.** Take the TFIM chain of Definition 3.3, with A the bond terms and B the field terms.

(a) Show that $[Z_1Z_2, X_1] = 2i\,Y_1Z_2$, and that Z_iZ_{i+1} commutes with X_j for $j \neq i, i+1$.

(b) Show that $\|[A,B]\| \leq 4Jh(n-1)$.

Hint: Write $[A,B]$ as a sum of Pauli strings, and bound its norm by its Pauli 1-norm.

Exercise 3.6 (The cost of a step on a device)**.**

(a) How many CZs does one Trotter step of the *TFIM* on an $L \times L$ square lattice cost

on a device with a square coupling graph?

(b) Place a 3×3 lattice on a line of 9 qubits, along a path like that of Figure 3.2 (c). Which bonds join qubits that are not neighbours on the line, and how far apart are they?

(c) Implement each such bond by moving one of its qubits next to the other with SWAPs, applying R_{ZZ}, and moving it back. Without cancelling any gates, how many CZs does one step cost now?

Hint: A SWAP is three CNOTs (Figure 1.11), and each CNOT is one CZ with single-qubit gates (Exercise 1.6).

3.5.4 Large steps: Floquet circuits

What does the circuit do when δ is not small? Each step is still a unitary, and every unitary can be written as $e^{-iH_F\delta}$ for some Hermitian H_F, by taking the logarithms of its eigenvalues. The circuit therefore follows the exact evolution under the Hamiltonian H_F, seen at the times $t = \delta, 2\delta, \ldots$. Expanding $e^{-iH_F\delta}$ to second order in δ and comparing with Equation 3.8 gives

$$H_F = A + B + \frac{i\delta}{2}[A, B] + O(\delta^2), \tag{3.10}$$

which is close to H only for small δ. For small δ, this gives a second way to read the Trotter error: the circuit evolves exactly, but under H_F instead of H. Two Hamiltonians that differ by $\|H_F - H\| \approx (\delta/2)\|[A, B]\|$ give evolutions that drift apart by at most $t\,\|H_F - H\|$ in a time t, which for $\delta = t/r$ is the bound of Equation 3.9. At large δ, the circuit therefore simulates a different Hamiltonian: it costs the same per step, but its dynamics can be completely different from that of H. A circuit that repeats one layer of gates is called a *Floquet circuit*, after the mathematician Gaston Floquet, whose theory describes systems that are driven periodically in time, such as atoms under a pulsed laser. The circuit applies A and B in turn, over and over, and so acts like such a periodic drive.

> *Example.* Figure 3.5 shows both regimes for a *TFIM* chain of 6 qubits that starts in $|0\cdots0\rangle$. Panel (a) shows the magnetisation. With steps of 0.25, it follows the exact curve. With steps of 1, it follows a different one, for example -0.42 instead of 0.20 at $t = 1$. Panel (b) shows the *fidelity* of the circuit's state with the exact state (Definition 2.9). At steps of 0.5, the magnetisation stays within 0.1 of the exact curve, while the fidelity drops to values between 0.37 and 0.71 after the first step.

The fidelity compares the whole state, and small errors in many parts of it multiply, so local observables such as the magnetisation are often much more accurate than the state as a whole.

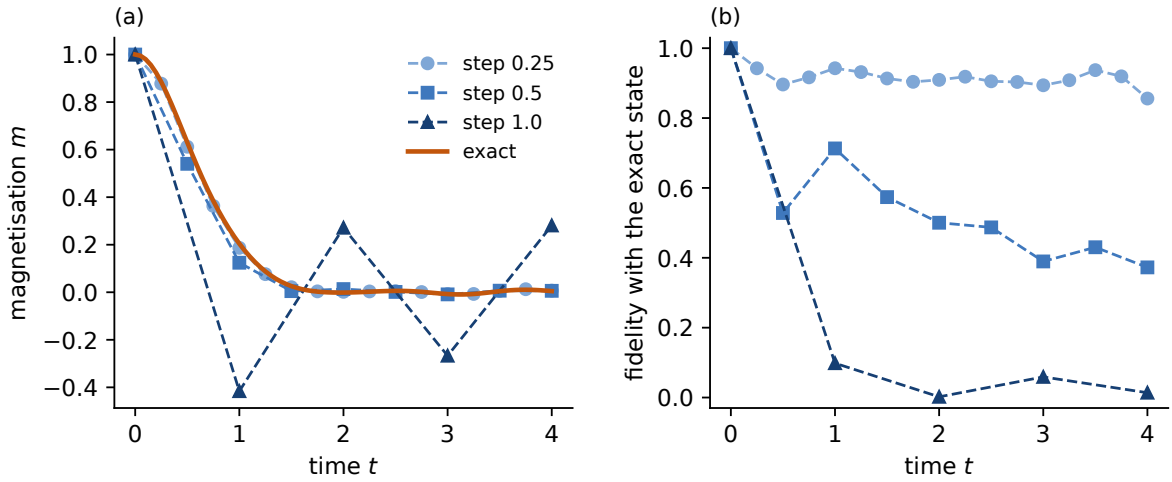


Figure 3.5: A TFIM chain of 6 qubits ($J = h = 1$) evolved from $|0\cdots0\rangle$, exactly (orange) and with Trotter steps of 0.25, 0.5 and 1 (blue, light to dark; the same markers in both panels). (a) The magnetisation; (b) the fidelity of the circuit's state with the exact state.

3.6 Observables after time evolution

Once the circuit has prepared $e^{-iHt}|\psi(0)\rangle$, we measure it. Three quantities are common, and the shots of the Z setting give all of them:

- the magnetisation m of Section 3.4;
- *correlation functions* $\langle Z_i Z_j\rangle$, which show how much the qubits i and j agree, and how this changes with their distance;
- the return probability $|\langle\psi(0)|\, e^{-iHt}\,|\psi(0)\rangle|^2$, which for $|\psi(0)\rangle = |0\cdots0\rangle$ is the fraction of shots that give all zeros; it decreases quickly with n, so large systems need many shots.

How fast does a change spread through a system?

> *Example.* Figure 3.6 shows a Heisenberg chain of 21 qubits, with $J = 1$, that starts in $|0\cdots0\rangle$ with the middle qubit flipped, evolved with Trotter steps of 0.2. Each bond gate conserves the number of flipped qubits, as the Hamiltonian does (Section 3.3), so there is always exactly one flipped qubit, and $(1 - \langle Z_i\rangle)/2$ is the probability of finding it on qubit i. Since $|0\cdots0\rangle$ is an eigenstate, the qubits that the flip has not reached show no change at all, so the spreading can be read off directly. Each bond gate partly swaps its two qubits (Exercise 3.1), and the flip spreads out from the middle.

The figure shows two cones, with different causes. The dashed lines are the *light cone* of the circuit. Since the Hamiltonian is geometrically local (Definition 3.2), each gate acts on two neighbouring qubits, and each step has two layers of bond gates, so after k steps the flip can

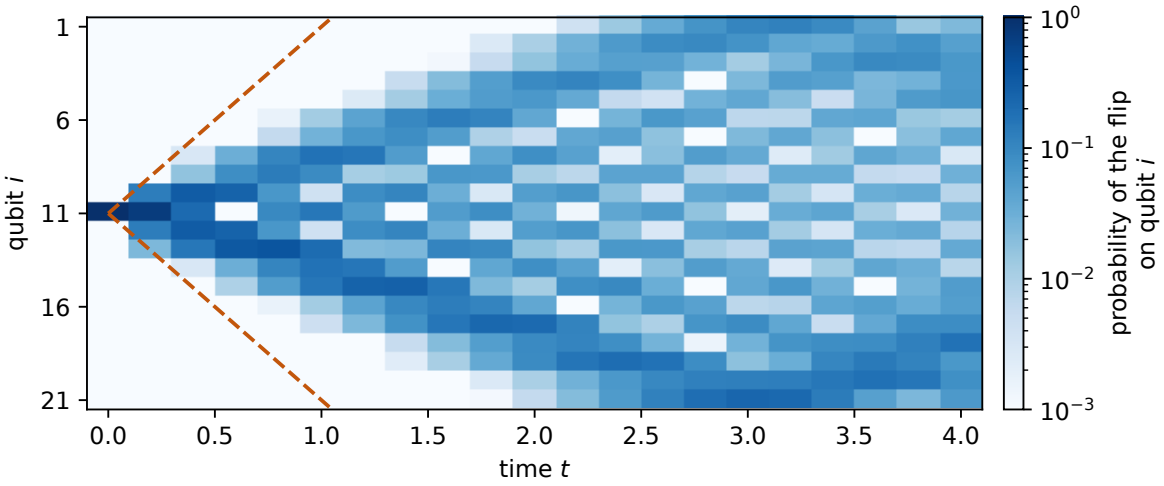


Figure 3.6: Light cones in a Heisenberg chain of 21 qubits ($J = 1$) that starts in $|0 \cdots 0\rangle$ with qubit 11 flipped, evolved with Trotter steps of 0.2. The colour is the probability of finding the flip on qubit i, on a logarithmic scale; white is 10^{-3} or less. Dashed: the light cone of the circuit, 2 qubits per side per step.

have reached at most $2k$ qubits on each side. Outside the dashed lines, the probability is exactly zero, up to rounding errors of order 10^{-15} in the simulation. This is the cone along which errors spread in Figure 2.4, and it holds for every circuit with this layout, whatever its angles. Read backwards, it also means that a local observable after k steps depends only on the qubits inside its cone, so that short circuits can be checked by simulating only those qubits (Exercise 3.8).

The coloured region is narrower: it is the cone of the Hamiltonian itself. In a geometrically local Hamiltonian, information spreads at a finite speed, and far away from the flip nothing changes for a while, up to tails that decrease exponentially with the distance. In the Heisenberg chain of the figure, the flip travels about $4J$ qubits per unit time, much slower than the $2/\delta = 10$ qubits per unit time of the circuit cone. When it reaches the ends of the chain, around $t = 2$, it is reflected, and soon it is spread over the whole chain. A circuit can only follow the physics while its cone is wider than that of the Hamiltonian, which is one more reason for small steps.

3.7 Trotter error against gate noise

More Trotter steps reduce the Trotter error, but each step adds gates, and with them noise. How many steps give the state closest to the exact one? We measure the distance with the fidelity of Definition 2.9, and estimate both errors with simple models.

- *Trotter error.* By Equation 3.9, the state of the circuit differs from the exact one by a vector of length at most at^2/r, with $a = \|[A, B]\|/2$. For two pure states, the infidelity $1-F$ is at most the square of this length, so the Trotter error contributes at most $(at^2/r)^2$.
- *Gate noise.* Let $\varepsilon_{\text{step}}$ be the sum of the Pauli errors of the gates of one step. By Equation 2.12, r steps run without an error with probability about $1 - r\varepsilon_{\text{step}}$, which approximates the fidelity with the noiseless state.

For small errors, the two infidelities add up,

$$1 - F \approx \left(\frac{at^2}{r}\right)^2 + r\,\varepsilon_{\text{step}}. \tag{3.11}$$

The first term falls with r, and the second grows. Setting the derivative with respect to r to zero gives the best number of steps,

$$r^* = \left(\frac{2a^2t^4}{\varepsilon_{\text{step}}}\right)^{1/3}. \tag{3.12}$$

Example. Take a *TFIM* chain of 4 qubits, $t = 1$ and $J = h = 1$. The bound gives $a = 2Jh(n-1) = 6$. For the noise, we take the depolarising *noise model* of Tutorial 2 with the values typical of current devices quoted there, $p = 0.1\%$ after each single-qubit gate and 0.5% after each CZ, which are Pauli errors of $3p/4$ and $15p/16$ (Definition 2.4 and Definition 2.8). As in Tutorial 2, the circuit is compiled into PRX and CZ gates without simplification, and every single-qubit gate, Z rotations included, is a noisy pulse. One step then has 37 single-qubit gates and 6 CZs, so $\varepsilon_{\text{step}} \approx 0.056$, and Equation 3.12 gives $r^* \approx 11$. A simulation of the noisy circuit (Figure 3.7) finds the best fidelity, 0.82, at 3 steps instead. The bound is pessimistic for three reasons: the exact $\|[A, B]\|$ is smaller than our estimate (8.9 instead of 12 here), the bound holds for the worst initial state, and it assumes that the errors of all steps add up. The value of a that fits the simulation without noise, $a \approx 0.70$, gives $r^* \approx 2.6$, close to the simulated optimum. The model therefore gives the shape of the trade-off and how it scales, while the best number of steps in practice is found by simulating small instances. For local observables, which are more accurate than the state (Figure 3.5), fewer steps can be enough.

Exercise 3.7 (How many steps?)**.** A circuit evolves a system for a time $t = 1$, with $a = 1$ in Equation 3.11, on a device with $\varepsilon_{\text{step}} = 0.01$.

(a) Compute r^* and the two terms of the infidelity at r^*. How do they compare? Is this a coincidence?

(b) The device improves tenfold, to $\varepsilon_{\text{step}} = 0.001$. By which factors do r^* and the smallest infidelity change?

(c) Back at $\varepsilon_{\text{step}} = 0.01$, suppose we use a bound that overestimates a ninefold, as in the example of the text. How many steps do we choose, and what is the infidelity of the

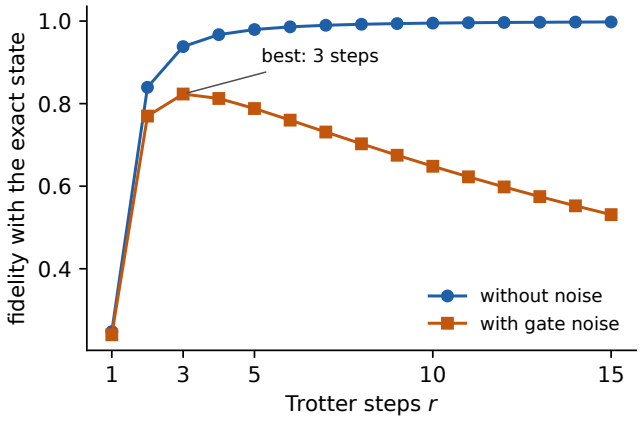


Figure 3.7: The fidelity with the exact state after a time $t = 1$, for a TFIM chain of 4 qubits ($J = h = 1$), against the number of Trotter steps r: without noise (blue) and with depolarising noise after every native gate (orange; $p = 0.1\%$ per single-qubit gate, 0.5% per CZ). With noise, the fidelity is highest at 3 steps.

circuit, with the true $a = 1$?

3.8 More exercises

Exercise 3.8 (Light cones and classical simulation)**.**

(a) In the Trotter circuit of Figure 3.4 on a long chain, on which qubits of the initial state can $\langle Z_i \rangle$ depend after k steps?

(b) On a square lattice, each step has four layers of bond gates, two along the rows and two along the columns. Which qubits can $\langle Z_i \rangle$ depend on after k steps, and how many are they?

(c) A laptop can simulate about 30 qubits exactly. For how many steps can it compute $\langle Z_i \rangle$ exactly on a chain of 1000 qubits, and on a square lattice of 1000 qubits?

Exercise 3.9 (The energy of a noisy state)**.** A circuit prepares a state of the *TFIM* chain of n qubits, but each qubit undergoes local depolarising noise with parameter p (Definition 2.7) just before the measurement. Write E_A and E_B for the ideal expectation values of the bond terms and of the field terms.

(a) Which energy does the measurement give on average, in terms of E_A, E_B and p?

(b) For $n = 10$, $p = 1\%$, $E_A = -6$ and $E_B = -5$, compute the *bias*. Compare it with the standard error of 0.1 of Section 3.4.1. Do more shots help?

(c) If p is known, can the bias be removed by dividing the estimate by a single factor, as in Section 2.7.2? How else?

3.9 Further reading

de Wolf (2026, ch. 9) treats Hamiltonian simulation from the algorithmic side, including methods that scale better than the Trotter formula. Childs et al. (2021) analyse Trotter errors in detail. Tilly et al. (2022, sec. 5) review measurement strategies for sums of Pauli strings, in the context of the variational methods of the next chapters: among them, grouping strings that commute in the ordinary sense, measured together after an entangling circuit, and *classical shadows*, which measure in random bases.

A Glossary

1/f noise Noise in which slow fluctuations are much stronger than fast ones, in inverse proportion to their frequency. It typically comes from many defects, each switching at its own rate.

amplitude damping The channel that describes relaxation: a qubit in the excited state decays to the ground state with probability gamma, which grows with time as one minus exp(-t/T1).

average gate fidelity The state fidelity between the output of a noisy gate and the ideal output, averaged over all pure input states. One minus it is the error per gate.

bias The difference between the average of an estimator and the true value. Noise biases estimates of expectation values; more shots do not reduce the bias.

bit flip The Pauli channel that applies X with some probability p: it flips the bit of a qubit. Readout errors with equal error rates act as a bit flip just before the measurement.

Bloch vector The three expectation values of X, Y and Z of a qubit, which fix its density matrix. It has length one for pure states and less for mixed states.

brick-wall circuit A circuit whose two-qubit gates act on alternating pairs of neighbouring qubits, first (1,2), (3,4), and so on, then (2,3), (4,5), and so on, like the bricks of a wall.

calibration Measuring and adjusting the control pulses of a device so that its gates do what they should. Providers publish calibration data: the error rates of each qubit and gate at the last calibration.

coherence The off-diagonal entry of a qubit's density matrix, which carries the relative phase of a superposition. Its size decays over the dephasing time T2*, or over the longer time T2 with an echo.

coherence time How long a qubit keeps its quantum state, in particular a superposition, before noise from its surroundings destroys it.

coherent error A systematic error in which a gate applies the wrong unitary, for example a rotation by a slightly wrong angle, in the same way every time.

completely positive A map of density matrices is completely positive if it still gives valid density matrices when it acts on part of a larger system. Quantum channels are exactly the completely positive maps that preserve the trace.

confusion matrix The table of probabilities of reading each outcome given each true state of the qubits. It describes readout errors.

correlated errors Errors that affect several qubits together, or that depend on what happens elsewhere in the circuit, such as crosstalk or bursts caused by cosmic rays.

correlation function An expectation value of a product of observables on two qubits, such as Z on qubit i times Z on qubit j, which shows how much the two qubits agree and how

this changes with their distance.

coupler A circuit element between two superconducting qubits through which they interact. A tunable coupler can switch the interaction on and off.

coupling graph Graph whose vertices are qubits and whose edges are the pairs on which a native two-qubit gate is available.

cross-resonance A two-qubit gate mechanism for superconducting qubits with fixed couplings: one qubit is driven at the frequency of its neighbour.

crosstalk An unwanted effect of an operation on one qubit on other qubits, for example through couplings that are never fully switched off.

density matrix The general description of a quantum state, including states that are uncertain mixtures of pure states: a positive semidefinite matrix with trace one. Expectation values are traces of the observable times the density matrix.

dephasing Loss of the relative phase of a superposition without loss of energy, for example because the qubit frequency is shifted by an unknown amount. The measured times T2* and T2 include both dephasing and relaxation; the time of pure dephasing alone is called T .

dephasing channel The Pauli channel that applies Z with some probability p, also called the phase-flip channel. It keeps the probabilities of 0 and 1 and shrinks the coherence. An unknown phase, for example from an unknown shift of the qubit frequency, acts as a dephasing channel.

depolarising channel On one qubit, the Pauli channel that applies X, Y or Z, each with probability p/4. It is the same as replacing the state by the maximally mixed state with probability p, and it shrinks the Bloch ball uniformly. On many qubits, it can be local or global.

depth The number of layers of gates in a circuit that must run one after another. Gates on different qubits can share a layer.

digital quantum computer A quantum computer that runs circuits: sequences of gates from a fixed universal set, applied to qubits and followed by measurements. Also called gate-based, as opposed to analog devices that imitate the time evolution of a physical system directly.

dilution refrigerator A refrigerator that reaches about 10 millikelvin, cold enough that heat no longer excites superconducting qubits.

dispersive readout Readout of a superconducting qubit through a resonator whose frequency depends slightly on the qubit state. A weak microwave probe tells the two states apart.

DiVincenzo criteria A checklist for building a quantum computer: scalable, well-characterised qubits; preparation of a known state; long coherence; a universal set of gates; measurement of individual qubits.

dynamical decoupling Repeated echo pulses during a long wait, which remove noise that is slow compared with the spacing of the pulses. It suppresses noise on the device itself.

echo An X pulse halfway through a wait, which reverses the phase accumulated before it, so that phase errors that stay constant during the wait cancel. An echo experiment measures T2.

error mitigation Methods that reduce the effect of noise on estimated quantities without

correcting the errors, typically by running extra or modified circuits and combining their results. They cost extra shots.

estimator A rule that turns measured data into an estimate of a quantity, for example the sample mean of the outcomes as an estimate of an expectation value.

fidelity A number between 0 and 1 measuring how close a state or operation is to the intended one. The state fidelity is the probability that a state passes a test for the target state; the average gate fidelity averages it over all input states. Informally, one minus the probability of an error.

Floquet circuit A circuit that repeats the same layer of gates many times, like a system driven periodically in time. With large Trotter steps, it follows the exact evolution under a Hamiltonian that differs from the intended one.

geometrically local Hamiltonian A Hamiltonian on qubits placed on a chain or lattice, whose terms each act only on qubits within a fixed distance of each other, for example single qubits and pairs of neighbours. Every qubit appears in only a few terms, so the number of terms is proportional to the number of qubits. A stronger property than k-locality.

global depolarising channel The depolarising channel acting on all qubits at once: with probability p, the state of all qubits is replaced by the maximally mixed state. It shrinks every Pauli expectation value by the same factor, whatever its weight.

ground state The eigenvector of a Hamiltonian with the lowest eigenvalue, the state of lowest energy. A system that is cold enough settles in or near its ground state.

Hamiltonian The observable of the energy of a system, a Hermitian matrix. Its eigenvalues are the possible energies, and it also determines how the state of the system changes in time.

Heisenberg model A spin model with the coupling XX + YY + ZZ on every pair of neighbouring qubits. Each bond treats the three axes alike, and the number of qubits in the state 1 is conserved.

incoherent error A random error that differs from shot to shot and averages to a noisy channel. Random errors accumulate more slowly than coherent ones.

ion An atom with one or more electrons removed, so that it carries an electric charge and can be held by electric fields.

ion trap A device that holds ions in vacuum with electric fields, typically lined up in a chain a few micrometres apart.

Josephson junction A thin insulating barrier between two superconductors. In a circuit it makes the energy levels unevenly spaced, which is what turns the circuit into a usable qubit.

k-local Hamiltonian A Hamiltonian that is a sum of terms that each act on at most k qubits, with k fixed as the number of qubits grows. As a sum of Pauli strings, it has polynomially many terms instead of exponentially many. Often simply called a local Hamiltonian.

Kraus operators A set of matrices that defines a quantum channel: the channel maps a density matrix to the sum of each Kraus operator times the density matrix times its adjoint. The same channel can have different sets of Kraus operators.

laser cooling Slowing the motion of atoms or ions with laser light until they are almost at rest.

leakage An error in which a qubit leaves its two qubit levels, for example when a transmon is pushed into its third level.

light cone The qubits that an operation, an error or a change can influence after a given number of circuit layers or a given time. In a circuit of nearest-neighbour gates, it grows by a fixed number of qubits per layer; a local Hamiltonian also spreads changes at a finite speed.

local depolarising noise Depolarising noise that acts on each qubit separately and independently. It shrinks the expectation value of a Pauli string by one factor for each qubit on which the string acts, so strings of higher weight shrink faster.

locality The principle that physical interactions involve only a few particles at a time, usually ones close to each other in space. It is why the Hamiltonians of physical systems are local, with polynomially many terms instead of exponentially many.

logical qubit A qubit of information encoded in many physical qubits by an error-correcting code. Its error rate can be much lower than that of the physical qubits.

magnetisation The average over the qubits of the expectation value of Z: the average alignment of the spins along z.

Markovian noise Noise without memory: the noise after each operation does not depend on what happened before. The model in which every gate is followed by a fixed channel assumes it.

mean squared error The average of the squared difference between an estimate and the true value. It equals the bias squared plus the variance.

measurement setting A choice of basis (X, Y or Z) for every qubit, made with single-qubit rotations before the measurement. Each shot of a setting gives a value of every Pauli string whose factors agree with it.

mixed state A quantum state that is not pure: a probabilistic mixture of pure states, described by a density matrix. On one qubit, its Bloch vector lies inside the Bloch sphere.

Mølmer-Sørensen gate The standard two-qubit gate of trapped ions. Laser beams couple two ions to the shared vibrations of the chain, producing an entangling rotation of any chosen angle.

native gate A gate the hardware implements directly with a calibrated control pulse. Every other gate must be compiled into native gates.

NISQ Noisy intermediate-scale quantum: devices with tens to thousands of qubits whose errors are not corrected, so noise limits the size of the circuits they can run.

noise model A description of the noise of a device, gate by gate and qubit by qubit, used to simulate noisy circuits, for example built from the calibration data of the device.

operator norm The largest factor by which a matrix can stretch the length of a vector. For a Hermitian matrix, it is the largest absolute value of its eigenvalues.

optical tweezers Tightly focused laser beams that hold single neutral atoms in place and can move them.

Pauli 1-norm The sum of the absolute values of the coefficients of an operator written as a sum of Pauli strings, often written lambda in the literature. It sets the worst-case number of shots needed to estimate the energy, and it bounds the operator norm.

Pauli channel A channel that applies a random Pauli error, chosen with fixed probabilities. It multiplies the expectation value of each Pauli string by its own factor.

Pauli string A tensor product of single-qubit Pauli matrices (I, X, Y or Z), one for each qubit,

such as ZZ or XIY. Hermitian, and squares to the identity.

phase-flip channel Another name for the dephasing channel, the Pauli channel that applies Z with some probability p. The name describes its Kraus form; physically, dephasing is usually a gradual loss of information about the phase.

phonon A quantum of vibration. In an ion trap, phonons are the quanta of the vibrations that the whole chain of ions shares; in a solid, such as a qubit chip, they are the quanta of the vibrations of the crystal.

phonon bus The shared vibrations of a chain of trapped ions, used as a common channel through which any two ions can interact in a two-qubit gate.

physical qubit A qubit as implemented in hardware, for example one transmon or one ion, with all its imperfections.

pulse A short burst of microwave or laser radiation at a chosen frequency, switched on and off smoothly, used to drive a qubit. Its frequency selects which qubit responds; its area and phase set the rotation.

pure dephasing The part of dephasing that does not come from relaxation, described by a dephasing channel whose strength grows with time, with characteristic time T . It cannot be measured on its own, since relaxation also reduces the coherence.

purity The trace of the square of a density matrix: 1 for pure states, smaller for mixed states, and smallest, one over 2 to the n, for the maximally mixed state of n qubits (1/2 for a uniformly random qubit).

QCCD Quantum charge-coupled device: a trapped-ion architecture in which ions are moved between zones of the trap, so that any two ions can be brought together for a gate.

quantum advantage A task that a quantum computer solves better than any classical method, faster, more cheaply or more accurately. For a task of a given size, it is a claim against the best classical methods known, and it is lost if better ones are found.

quantum channel The general description of how a quantum state changes, including noise: a linear map of density matrices that can be written with Kraus operators. Unitary gates are the channels with a single Kraus operator.

quantum dot A tiny region in a semiconductor that can trap single electrons.

quantum error correction Encoding quantum information redundantly in many physical qubits, so that errors can be detected and corrected without disturbing the encoded information.

qubit-wise commuting Two Pauli strings are qubit-wise commuting if, on every qubit, their factors are equal or one of them is the identity. Such strings can be measured from the same shots, in one measurement setting.

Ramsey experiment A measurement of dephasing: a pulse puts the qubit in a superposition, the qubit waits, and a second pulse turns the remaining coherence into a measurable population. It measures T2*.

random telegraph noise Noise from a defect that switches at random between two states, making the qubit frequency jump back and forth.

randomised benchmarking A method to measure the average error per gate: random gate sequences of growing length, which ideally return to the initial state, are run, and the decay of the success probability with the length gives the error per gate.

relaxation Loss of energy by a qubit: the excited state decays to the ground state. Its characteristic time is called T1, and the amplitude-damping channel describes it.

resonance Radiation drives a transition between two energy levels only when its frequency matches their energy difference divided by Planck's constant. This is how individual qubits are addressed.

Rydberg state A state of an atom in which one electron is excited very far from the nucleus. Atoms in Rydberg states interact strongly over micrometre distances.

Schrödinger equation The equation that relates states to the Hamiltonian. In its time-independent form, it is the eigenvalue equation of the Hamiltonian; in its time-dependent form, it says how a state evolves, and its solution applies the unitary exp(-iHt).

scrambling A property of circuits, such as deep circuits of random gates, that spread every error over all the qubits, so that any error leaves the output close to random. For such circuits, global depolarising noise is a good model of local noise.

shot One run of a circuit: prepare the qubits, apply the gates, measure. Returns one bit string.

SPAM errors State preparation and measurement errors, characterised together because a measurement right after a preparation cannot tell them apart.

spin An intrinsic two-level property of an electron (spin-1/2: up or down), which can serve as a qubit.

state fidelity For a pure target state, the probability that a state passes a test for the target, that is, the expectation value of the projector onto the target.

SWAP A two-qubit gate that exchanges the states of two qubits. It costs three CNOTs.

T2* The dephasing time: the time over which the coherence of a qubit decays in a Ramsey experiment. It is shorter than T2, measured with an echo, because it also includes slow noise that the echo removes.

tensor network A classical description of a quantum state as a network of small arrays, one per qubit, whose size grows with the entanglement. Efficient for weakly entangled states, such as those of a chain evolved for a short time.

threshold The physical error rate per operation below which error correction helps: below it, larger codes give lower logical error rates.

transmon The most common superconducting qubit: a small circuit with a Josephson junction, whose two lowest energy levels form the qubit.

transpilation Compilation of a circuit into the native gates and coupling graph of a device, including inserting SWAPs and simplifying.

transverse-field Ising model A spin model with a ZZ coupling on every pair of neighbouring qubits, which favours aligned neighbours, and a field along x on every qubit, which favours the state plus. Abbreviated TFIM.

Trotter formula An approximation of the evolution under a sum of terms by a product of evolutions under each group of terms in turn, repeated in many short Trotter steps. Its error, the Trotter error, comes from terms that do not commute and decreases as the number of steps grows.

twirling Inserting random Pauli gates around a gate, chosen so that they cancel in the ideal circuit. Averaged over the random choices, any error becomes a Pauli channel.

unital A channel is unital if it maps the maximally mixed state to itself; on one qubit, it keeps

the centre of the Bloch ball fixed. Pauli channels are unital; relaxation, described by amplitude damping, is not.

virtual Z gate A Z rotation implemented by shifting the phase of all later drive pulses instead of applying a pulse. It takes no time and adds no error.

weight The number of qubits on which a Pauli string acts with a Pauli matrix other than the identity. Under local noise, strings of higher weight lose their expectation value faster.

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